

Part I: Measurement Cost of Generator Selection in Fixed-State ADAPT-VQE

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Abstract

This report defines and compares three measurement strategies for the generator-selection step in gradient-based ADAPT-VQE (adaptive derivative-assembled pseudo-Trotter variational quantum eigensolver): **M1: FC-UG**, fully commuting (FC) universal groups for static all-gradient estimation; **M2: BAI-FC-IG**, best-arm identification (BAI) using FC individual-gradient measurements; and **M3: BAI-FC-UG**, best-arm identification using fixed parent FC universal groups. A fourth construction, a *no-sharing baseline* that uses neither mechanism, completes the two-by-two the three methods span and makes each mechanism separately attributable.

Two classes of number are reported, and they are kept apart throughout. *Planning bounds* (Secs. 9–10) are oracle diagnostics: exact gradients and exact fragment variances are used to compute the shots each method would need, and the elimination rule reads exact values, so it cannot misfire. A *finite-shot surrogate* (Sec. 12) replaces the exact gradients with Gaussian estimates built from a finite number of shots and runs the same elimination rule on those. The surrogate samples estimators, not individual measurement outcomes, so it is not a shot-level simulation of a quantum device; the distinction and its consequences are set out in Sec. 7.

Results. On the planning bounds, M3 is cheaper than M1 in all nine cases, by factors of 1.1 to 192, and cheaper than M2 in eight of nine; neither ordering is a theorem, and small problems with a nearly tied runner-up exceed M1 by up to 9%. Against the no-sharing baseline, elimination alone is worth 1.3 to 253 and sharing alone only 1.1 to 3.4. The planning bound resolves every M3 arm to a common radius, but on LiH and H₂O the shared allocation resolves the leader far more tightly, and M3 eliminating on that actual precision costs only 0.52 to 0.74 of its bound. On those five rows the M2/M3 ratio widens to 1.19–3.42, M3 wins both stretched H₂O rows that the bound leaves unresolved under every grouping routine tested, and the two mechanisms compound rather than substitute (interaction 1.05–1.79); on H₄, and on every row at bound level, they substitute. Under the finite-shot surrogate both best-arm methods cost 2.2 to 3.0 times their bounds on a coarse radius schedule; near the exact limit that falls to 1.03–1.35 for M2 and 0.73–1.29 for M3, so the inflation is discretisation, not noise, and M3 beats M1 on all nine simulated rows. Modelling the within-context correlation and eliminating on pairwise radii cuts M3’s realised cost by a further 33 to 49% at no measured loss of accuracy. Every trial of the main finite-shot table returned the correct generator; the stated family-wise failure probability does not control the error rate, which stays below 2% even at $\delta = 0.99$.

Scope. Every number is a single-ADAPT-step diagnostic for one fixed state, and counts shots only: circuit depth, the number of distinct measurement settings, and state-preparation cost are excluded (Sec. 7). Part III symmetry filtering and tapering are not included in the Part I tables; Part III supplies a smaller input problem to the same measurement methods.

1 Problem statement and relation to previous work

Given a molecular Hamiltonian H , a current ADAPT-VQE state $|\psi\rangle$, and a unitary coupled-cluster singles-and-doubles (UCCSD) generator pool $\{G_i\}$, estimate the measurement cost required to identify a generator with the largest absolute gradient

$$i^* = \arg \max_i |g_i|, \quad g_i = \langle \psi | [H, G_i] | \psi \rangle. \quad (1)$$

The scientific question is whether using winner-identification statistics and shared fully commuting measurement contexts reduces the gradient-selection measurement cost relative to static all-gradient estimation.

The answer, in one sentence. Mostly yes: on oracle planning bounds at family-wise $\delta = 0.05$, combining both mechanisms (M3) costs 1.1 to 192 times less than static all-gradient estimation (M1) across the nine cases studied, and the same ordering holds on realised cost under a well-resolved schedule. The change of statistical target carries most of the saving, shared contexts being worth only 1.1 to 3.4 alone — but on the larger systems sharing also makes elimination fire earlier, by resolving the leader more tightly than required, and there the two compound.

Here g_i is the derivative of the energy with respect to the variational parameter of generator G_i at $\theta_i = 0$, in Hartree per radian; G_i is anti-Hermitian, so g_i is real. The generator pool and its labelling convention are fixed in Sec. 6.

Relation to previous work. ADAPT-VQE and the spin-orbital UCCSD pool used here are those of Ref. [1]. Two lines of work attack the selection-step measurement cost separately. Ref. [3] attacks it by *sharing*: it partitions the gradient observables into commuting sets using Hamiltonian terms as pivots, reducing the pool-gradient cost to a small multiple of one energy measurement, and Ref. [2] pursues a related reduction alongside symmetry considerations. Ref. [4] attacks it by *elimination*: it recasts generator selection as best-arm identification and applies successive elimination, discarding unpromising candidates early. Ref. [5] changes the selection criterion itself.

The two mechanisms have not, to our knowledge, been measured against each other on one cost criterion, and it is not obvious that they compose. This report supplies the missing comparison rather than a new algorithm. Its contributions are: the 2×2 factorial of sharing and elimination, completed by a no-sharing baseline so that each mechanism and their *interaction* are separately attributable (Sec. 10); a single shot criterion with no free schedule or precision parameter, applied identically to all four constructions; and the finding that, at the level of the planning bounds, the two mechanisms are substitutes rather than complements, with elimination carrying the result. M1 stands in for the sharing line, M2 for the elimination line, and M3 for their combination.

2 Fixed method names

The phrase “naive/no-sharing” is retired as a *method* name. A construction that shares nothing does appear below, as the fourth corner of the two-by-two that the three methods span; it is called the *no-sharing baseline* throughout, it is a diagnostic rather than a project method, and it is defined in Sec. 10.

M1: FC-UG, Fully Commuting Universal Groups. One global parent FC grouping of the union of all Pauli products in all gradient commutators; estimate the full gradient vector.

M2: BAI-FC-IG, BAI with FC Individual Gradients. Each generator gradient is a BAI arm, measured in an FC grouping of its own commutator. No cross-gradient sharing.

M3: BAI-FC-UG, BAI with FC Universal Groups. The same fixed parent contexts as M1, with BAI elimination, sampling only the parent contexts needed to separate surviving generators.

No-sharing baseline. A diagnostic, not a project method: each generator grouped separately as in M2, every one resolved to the common radius used by M1 (Sec. 10).

3 Object lineage

The qubit Hamiltonian is

$$H = \sum_p h_p P_p, \quad (2)$$

where P_p are Pauli products after Jordan–Wigner mapping. For each generator G_i , define the gradient operator

$$C_i = [H, G_i]. \quad (3)$$

The ADAPT gradient is the expectation value of this operator,

$$g_i = \langle C_i \rangle_\psi. \quad (4)$$

Expanding all gradient operators in one shared Pauli-coordinate system gives

$$C_i = \sum_\ell A_{i\ell} R_\ell, \quad \mu_\ell = \langle R_\ell \rangle_\psi, \quad \mathbf{g} = A\boldsymbol{\mu}. \quad (5)$$

The parent universal Pauli support at this ADAPT step is

$$\mathcal{B}_0 = \bigcup_i \text{supp}(C_i). \quad (6)$$

M1 and M3 use $\mathcal{B}_0$ as a shared measurement object. M2 does not; it groups each C_i separately.

4 The three algorithms

4.1 M1: FC-UG

M1 estimates the full gradient vector. It forms one global FC grouping of the parent support,

$$\mathcal{B}_0 \longrightarrow \mathcal{M}_1, \dots, \mathcal{M}_K, \quad (7)$$

where each $\mathcal{M}_\alpha$ is a fully commuting measurement context. Measurements of these contexts estimate the shared Pauli expectations μ_ℓ and reconstruct every gradient using Eq. (5).

The implementation uses deterministic greedy FC grouping of $\mathcal{B}_0$ (Sec. 6). It is a generic stand-in for the pool-wide grouping strategies of Refs. [3, 2] and is not claimed to reproduce either construction; Sec. 14 shows how much the choice of routine matters.

4.2 M2: BAI-FC-IG

M2 changes the statistical target from full-vector estimation to winner identification. Each candidate gradient is an arm in a best-arm-identification problem. Its own commutator support is grouped separately,

$$\text{supp}(C_i) \longrightarrow \mathcal{M}_{i,1}, \dots, \mathcal{M}_{i,K_i}. \quad (8)$$

Throughout, t indexes the BAI round and r denotes a confidence radius; the two are never the same symbol. BAI maintains confidence intervals for active gradients and eliminates a generator i at round t if

$$|\hat{g}_i| + r_i < \max_{j \in \mathcal{A}_t} (|\hat{g}_j| - r_j), \quad (9)$$

where $\mathcal{A}_t$ is the active generator set and $r_i = z_\delta \epsilon_i$ is the current confidence radius of arm i , with ϵ_i its one-standard-error target and z_δ the confidence factor of Eq. (20). All methods in this report use a single common radius $r_i = r$ across active arms; the rule is written with per-arm radii because the covariance-aware variant of Sec. 15 does not. M2 does not exploit universal cross-gradient sharing.

Implementation status. M2 is now implemented as its own module. Each run writes the individual-gradient FC grouping $\mathcal{M}_{i,k}$ for every generator, the oracle fragment standard deviation of every fragment, the per-arm shot cost, and the full BAI elimination history with confidence radii and active sets. The M2 column is therefore auditable; it remains an oracle planning diagnostic rather than a finite-shot result.

4.3 M3: BAI-FC-UG

M3 combines the mechanisms of M1 and M2. It starts with the same fixed parent universal FC contexts as M1,

$$\mathcal{M}_1, \dots, \mathcal{M}_K, \quad (10)$$

but BAI eliminates generator candidates,

$$\mathcal{A}_0 \supset \mathcal{A}_1 \supset \mathcal{A}_2 \supset \dots. \quad (11)$$

At BAI round t , the active Pauli support is

$$\mathcal{B}_t = \bigcup_{i \in \mathcal{A}_t} \text{supp}(C_i), \quad \mathcal{B}_t \subseteq \mathcal{B}_0. \quad (12)$$

The active measurement contexts are restricted parent contexts,

$$\mathcal{M}_\alpha^{(t)} = \mathcal{M}_\alpha \cap \mathcal{B}_t. \quad (13)$$

The parent contexts are not rebuilt during BAI. This nested construction keeps previous measurements compatible while allowing the algorithm to stop sampling information that only affects eliminated generators.

How M3's cost is aggregated. Shots taken in a context are reusable, so a context is charged once, at the largest number of shots any round ever demanded of it. Writing $n_\alpha^{(t)}$ for the allocation of Eq. (21) solved over the active set $\mathcal{A}_t$ at the radius of round t , the reported M3 cost is

$$M_{\text{M3}} = \sum_{\alpha} \max_t n_\alpha^{(t)}. \quad (14)$$

Each round's allocation is solved independently, without reference to shots already committed; the maximum is taken afterwards. This is the accounting a real run would get for free by keeping its data, and it is why the trajectory is fully determined by the finitely many rounds at which the active set changes.

5 Oracle shot proxy

All numbers in Secs. 9–10 are oracle planning estimates: the planner is given the exact g_i and $\sigma_{i\alpha}$, and nothing is learned from shots. *Oracle* and *noiseless* mean only that. They do not refer to a noiseless device, and state-preparation error is excluded from the model altogether (Sec. 7).

Sorted order and the gap. Order the pool by decreasing absolute gradient, $|g_{(1)}| \geq |g_{(2)}| \geq \dots$; the parenthesised index is a rank, never a generator index. Then

$$\text{gap} = |g_{(1)}| - |g_{(2)}|, \quad \Delta_i = |g_{(1)}| - |g_i| \geq 0, \quad (15)$$

so $\Delta_{(2)} = \text{gap}$ and $\Delta_{(1)} = 0$.

Shots and context-shots. One *shot* is one state preparation plus one measurement of one context: a single circuit execution. *Context-shot* is the same thing charged to the context it was spent in; every cost quoted is $\sum_{\alpha} n_{\alpha}$. The metric counts shots only — not circuit depth, not the number of *distinct* measurement settings, not classical post-processing (Sec. 7).

For one gradient operator written as a sum of independently measured FC fragments,

$$C_i = \sum_{\alpha=1}^{K_i} F_{i\alpha}, \quad (16)$$

with oracle fragment variances

$$\sigma_{i\alpha}^2 = \text{Var}_{\psi}(F_{i\alpha}) = \langle F_{i\alpha}^2 \rangle_{\psi} - \langle F_{i\alpha} \rangle_{\psi}^2, \quad (17)$$

the optimal independent-fragment allocation gives the approximate total number of shots needed to estimate g_i with one-standard-error target ϵ_i as

$$M_i(\epsilon_i) \approx \frac{\left(\sum_{\alpha=1}^{K_i} \sigma_{i\alpha} \right)^2}{\epsilon_i^2}. \quad (18)$$

Precision target and confidence radius. ϵ_i is a one-standard-error target on $\hat{g}_i$; a *confidence radius* is the half-width of the corresponding interval,

$$r_i = z_{\delta} \epsilon_i, \quad (19)$$

so a method asked for radius r must reach $\epsilon = r/z_{\delta}$, and ϵ_i in Eq. (18) is replaced by r_i/z_{δ} . The confidence factor is the two-sided normal quantile with a Bonferroni correction over the whole pool,

$$z_{\delta} = \Phi^{-1} \left(1 - \frac{\delta}{2N} \right), \quad N = |\{G_i\}|, \quad (20)$$

with N the *full* pool size — 26, 92, 140 — not the non-zero count and not the active set, and no correction over rounds. At $\delta = 0.05$ this is $z_{\delta} = 3.10, 3.46$ and 3.57 , applied identically to all four constructions so that it cancels from every planning-bound ratio (Sec. 13). $\delta = 0.05$ is the conventional level; Bonferroni is chosen as the simplest family-wise correction valid under arbitrary dependence between the arms, which matters here because M1 and M3 read several arms off the same shots; Holm’s step-down procedure is equally valid and uniformly more powerful, and would be the better default. The price is conservatism, and Sec. 12.4 measures it: the construction does not control the realised selection-error rate, so it is a convention rather than a guarantee.

Eq. (18) is applied separately inside each gradient arm, because M2 measures each arm in its own contexts. M1 and M3 instead share contexts between generators, so the allocation over contexts is not separable and is obtained by solving the convex programme

$$\min_{n_{\alpha} \geq 0} \sum_{\alpha} n_{\alpha} \quad \text{subject to} \quad \text{Var}(\hat{g}_i) = \sum_{\alpha} \frac{\sigma_{i\alpha}^2}{n_{\alpha}} \leq \epsilon^2 \quad \text{for every active } i, \quad (21)$$

where n_{α} is the number of shots taken in context α and $\text{Var}(\hat{g}_i) = \sum_{\alpha} \sigma_{i\alpha}^2 / n_{\alpha}$ is the *reconstruction variance* of gradient i under that allocation: the variance of the estimate obtained by reading $\hat{g}_i$ off the shared contexts through Eq. (5). The quantity $\sum_{\alpha} \sigma_{i\alpha}^2$ that appears repeatedly below is its value at one shot per context, and serves as a scale-free measure of how expensive generator i is to reconstruct from the parent grouping. Its stationarity condition is $n_{\alpha} = \sqrt{\sum_i \lambda_i \sigma_{i\alpha}^2}$ for multipliers $\lambda_i \geq 0$, solved by a damped multiplicative fixed-point iteration (damping 0.5, at most 500 iterations, stopping when the least-loaded constraint is within 10^{-9} of the tightest) and rescaled so the tightest constraint holds exactly. The rescaling makes the result feasible whatever the iteration converged to, so

the stopping rule affects optimality and never validity. It does affect the last digits: rebuilt in a second numerical environment, M1 and M2 reproduce exactly on every row, while M3, whose total collects many allocations, moves by 0.1 to 0.6% on four rows (always downward). Table 3 quotes the original run; sections that recompute M3 quote the recomputation, and no ratio moves by more than 0.02. For a single gradient it reduces exactly to Eq. (18), and it matches a general convex solver on the instances tested (Sec. 19).

Shot counts are not rounded. n_α is real-valued with no per-context minimum; the per-arm closed form of Eq. (29) is rounded up per arm. A floor of one shot per context would add at most K shots, at most 0.2% of any entry in Table 3 (worst case: 53 contexts under M3’s 33,432 on H_4 side 2.0 CISD), changing quoted digits but no ratio.

The schedule-free limit. As the common radius r decreases continuously, generator i leaves the active set the moment $2r$ falls below Δ_i , so it is resolved to exactly $\Delta_i/2$ and the leader survives to gap/2. Measurements are reusable, so an arm leaving at radius r has paid for radius r and no more, and a shared context has paid the largest requirement it ever saw. The active set changes only at the distinct values of Δ_i , so evaluating the allocation there determines the whole trajectory; for M2 it collapses to Eq. (29). This removes the free shrink factor (Sec. 13).

Because these formulae depend on the exact grouping and fragment variances, every production run saves the FC groups, the fragment variances and the BAI histories used to generate each table entry.

6 Computational details

6.1 Molecular input

All cases use the minimal STO-3G basis, charge 0, and a closed-shell singlet restricted Hartree–Fock reference, with no frozen core and no qubit tapering. Water is placed in the xz plane with the oxygen at the origin and the bond angle held at 104.52° while $R(\text{OH})$ is varied, so the stretch is symmetric. The set is chosen to vary pool size at fixed basis and method — 26, 92 and 140 generators on 8, 12 and 14 qubits — while staying small enough that the exact FCI energy and exact fragment variances are available, since every number here depends on having them. The stretched geometries (LiH at 3.0 Å, H_4 at side 2.0 Å, H_2O at 1.75 Å) put each system in a strongly correlated regime. On H_4 and H_2O this also crowds the leading gradients together and makes selection hard; on LiH it does not, its gap of 0.096 being among the widest here, and with no LiH equilibrium row there is nothing to compare it against. The two H_2O geometries come closest to isolating the effect at fixed pool size, but not cleanly: the orbital reordering documented in Sec. 8 means their Pauli-level structures differ. Four systems cannot separate size from geometry and state, and no claim in this report about a trend with size should be read as established.

Geometries, in Å: LiH is Li at the origin and H at $(0, 0, 3.0)$, 6 spatial orbitals. H_4 is a planar square in the xy plane, H at $(0, 0, 0)$, $(s, 0, 0)$, $(0, s, 0)$, $(s, s, 0)$ with $s = 1.0$ and 2.0 , 4 orbitals. H_2O has O at the origin and H at $(\pm R \sin \frac{\theta}{2}, 0, R \cos \frac{\theta}{2})$ with $\theta = 104.52^\circ$ and $R = 0.9572$ and 1.75 , 7 orbitals.

Two states are used per geometry. **HF** is the closed-shell Hartree–Fock determinant. **CISD** (configuration interaction with singles and doubles) is the lowest eigenvector of the qubit Hamiltonian restricted to the determinants within double excitation of that reference, in the target (N_α, N_β) sector, normalised to unity and embedded in the full 2^n -dimensional space. The CISD state is used as a correlated proxy for a later ADAPT state: it is a correlated wavefunction in the same sector, cheap to construct exactly, and it shifts the gradient spectrum in the direction a converging ADAPT run does — the top gradient falls by a factor of 2.1 to 44 (the $|g_{(1)}|$ values are listed with Table 3), which is what makes selection expensive late in a trajectory. That is the whole of the justification. It is not a saved trajectory snapshot, and no evidence is offered that its gradient *spectrum* resembles one; testing that needs trajectory data (Sec. 18). LiH is run with the HF reference only; there is no reason of principle for the omission.

6.2 Generator pool, conventions and labels

Spin orbitals are interleaved: spatial orbital p gives spin orbitals $2p$ (α) and $2p + 1$ (β), zero-based. The computational basis is indexed big-endian, matching the sparse-operator convention of the linear-algebra backend. The Jordan–Wigner transform is used throughout.

The generator universe is the standard spin-orbital UCCSD pool. Each generator is anti-Hermitian,

$$G_i = \mathcal{JW}(\tau_i - \tau_i^\dagger), \quad \tau_i^S = a_a^\dagger a_i, \quad \tau_i^D = a_a^\dagger a_b^\dagger a_j a_i, \quad (22)$$

so that $g_i = \langle [H, G_i] \rangle_\psi$ of Eq. (1) is real and equals $\partial E / \partial \theta_i$ at $\theta_i = 0$ for the parameterisation $e^{\theta_i G_i}$. Singles run over occupied–virtual pairs of the same spin; doubles conserve N_α and N_β separately. The pool therefore conserves electron number and S_z but not generally S^2 . Ordering is deterministic: all singles in ascending (i, a) , then all doubles in ascending (i, j, a, b) . The labels used in the tables are exactly this: S $i \rightarrow a$ for a single and D $i, j \rightarrow$

a,b for a double, with zero-based interleaved spin-orbital indices. The resulting pool sizes are 26, 92 and 140 for H_4 , LiH and H_2O . This is the pool of the original ADAPT-VQE construction [1], chosen so that the observable structure in Sec. 8 is the standard one; qubit and spin-adapted pools give different supports and different pool sizes, and the comparison has not been repeated for them.

6.3 Fully commuting grouping routine

FC group counts are routine-dependent, and two rows are comparable only if produced by the same deterministic routine with the same Pauli ordering, tie-breaking rule and seed. Every row in the main tables uses one routine: deterministic greedy first-fit grouping, with Pauli strings sorted by decreasing weight (the number of non-identity factors) and ties broken by ascending lexicographic order of the string read left to right from qubit 0, under the letter order $\text{I} < \text{X} < \text{Y} < \text{Z}$, with each candidate joining the lowest-numbered existing group that accepts it and opening a new group otherwise. No random seed is used. Sec. 14 varies this convention deliberately and reports the spread.

Fully commuting grouping is chosen over qubit-wise commuting because it admits at least as many terms per context, so it tends to reduce the number of contexts and with it the quantity this report counts — though not necessarily, since positive covariance between terms inside a context raises that context’s variance. The cost of that choice is a Clifford diagonalisation circuit per context, which the shot metric does not price (Sec. 7); a qubit-wise-commuting comparison would trade more contexts for cheaper circuits and is the natural companion study once circuits are priced.

6.4 Validation gate

Every case is checked before any gradient is computed: the Jordan–Wigner (JW) Hamiltonian must reproduce the PySCF restricted Hartree–Fock (RHF) energy on the Hartree–Fock (HF) determinant and the PySCF full configuration interaction (FCI) energy as its lowest eigenvalue in the (N_α, N_β) sector, both to 10^{-8} Ha. All nine cases pass, with a maximum observed error of 4×10^{-15} Ha. A case that fails the gate aborts rather than producing numbers.

6.5 Non-zero gradient threshold

Gradients that vanish by symmetry appear at the 10^{-8} numerical noise floor of the integral and SCF pipeline, while the smallest physically non-zero gradient in these cases is of order 10^{-3} . The counts below use a threshold of 10^{-6} , which lies on the plateau between the two. The threshold is not innocuous at tighter values: for LiH the count is 26 at 10^{-6} and 34 at 10^{-9} , and every run records the count at 10^{-4} , 10^{-6} , 10^{-8} and 10^{-9} so the choice stays visible. **The threshold is used for reporting only.** No method filters the pool by it: all four constructions carry every one of the N generators as an arm, and N in Eq. (20) is the full pool size. The one place the distinction is used analytically is the diagnostic in Sec. 9.2, which asks what M1 would cost if it were given the non-zero set for free.

7 What the model assumes and excludes

Statistical assumptions, in one place. (i) Estimators are Gaussian; the normal limit is assumed, not tested against shot-level sampling of measurement outcomes. (ii) Distinct measurement contexts are sampled independently. (iii) Within a context, the planning bounds and the main finite-shot tables treat different generators’ fragments as independent; this is exact for M2 and the baseline, which give every generator its own shots, and an approximation for M1 and M3. Sec. 15 removes it and reports what changes. (iv) State preparation is exact and noiseless, and there is no gate, readout or decoherence error anywhere in the model. (v) The union bound behind Eq. (20) is taken over the N pool members only; it is not taken over the sequence of elimination rounds, which in the exact limit is a continuum. (vi) Fragment variances are the exact $\text{Var}_\psi(F_{i\alpha})$, so no cost is charged for learning them. (vii) In the finite-shot surrogate the fluctuation of each context is carried as a running sum over the shots taken in it, so successive rounds are correlated as real accumulated data would be; what is not modelled is the cost of learning the variances, per (vi).

“System size” means the pool size N , ordering H_4 (26) < LiH (92) < H_2O (140). On these systems that agrees with qubit count (8, 12, 14) and $|\mathcal{B}_0|$ (1,276; 30,196; 81,566) but not with electron count. With four systems, and geometry and state varying with them, no claim here about a trend with size is established; the phrase labels the ordering.

What the cost metric leaves out. Circuit depth and two-qubit count for the diagonalising Cliffords, compilation and calibration of distinct measurement settings, and classical post-processing. The number of distinct settings differs sharply between the methods, against the individually grouped ones.

M2 and the baseline need 3.5 to 6.1 times as many distinct settings as M1 and M3 here. Nothing in this report prices that, and a metric that did could move the M2-versus-M3 comparison, which on shots alone is already within

Table 1: Distinct measurement settings each construction requires, a cost component the shot metric does not price. M1 and M3 share one parent grouping; M2 and the baseline group every generator separately, so their settings are the sum over arms and are not reused between arms.

Case	M1, M3: K	M2, baseline: $\sum_i K_i$	Largest single arm: $\max_i K_i$
LiH, $R = 3.0$ Å	980	5,744	101
H ₄ square, side 1.0 and 2.0 Å	53	183	10
H ₂ O eq., $R(\text{OH}) = 0.9572$ Å	2,366	13,235	159
H ₂ O stretch, $R(\text{OH}) = 1.75$ Å	2,356	14,263	162

a factor of two on most rows. The counts are structural properties of the groupings, not measured circuit costs.

8 Observable structure for all systems

Table 2: Global universal support used by M1: FC-UG and M3: BAI-FC-UG. The FC contexts are global parent contexts for $\mathcal{B}_0$, not per-gradient groupings, and all rows use the single greedy routine fixed in Sec. 6. Non-zero gradient counts use the 10^{-6} threshold. Distances in Å.

Case	State	Qubits	Pool	Universal terms	FC-UG contexts	Nonzero g_i
LiH, $R=3.0$	HF	12	92	30,196	980	26
H ₄ , side 1.0	HF	8	26	1,276	53	10
H ₄ , side 1.0	CISD	8	26	1,276	53	10
H ₄ , side 2.0	HF	8	26	1,276	53	10
H ₄ , side 2.0	CISD	8	26	1,276	53	10
H ₂ O eq., $R=0.9572$	HF	14	140	81,566	2,366	40
H ₂ O eq., $R=0.9572$	CISD	14	140	81,566	2,366	48
H ₂ O str., $R=1.75$	HF	14	140	81,566	2,356	40
H ₂ O str., $R=1.75$	CISD	14	140	81,566	2,356	48

All FC context counts come from the single routine fixed in Sec. 6.

Why the two H₂O geometries have the same support size but different context counts. The grouping routine is deterministic in the set of Pauli strings alone, so equal supports would have to give equal counts; 2,366 against 2,356 therefore means the supports are not equal. They are not. Both contain 81,566 products, and $\sum_i |\text{supp}(C_i)|$ is 314,720 in both, but their intersection is only 78,113: each geometry has 3,453 Pauli products the other lacks, a symmetric difference of 6,906, or 8.5% of each. The differing products are not numerical dust — their largest commutator coefficients reach 0.09 and 0.11, and their medians, 2.6×10^{-3} and 3.7×10^{-3} , are within a factor of two of the median over the shared products.

The cause is an orbital reordering. Both geometries have the same point group and the same multiset of orbital symmetries, but the orbital energies cross: the canonical order is $(A_1, A_1, B_2, A_1, B_1, A_1, B_2)$ at 0.9572 Å and $(A_1, A_1, B_1, B_2, A_1, A_1, B_2)$ at 1.75. The symmetry-allowed integrals are equal in number, which is why $|\mathcal{B}_0|$ and $\sum_i |\text{supp}(C_i)|$ agree exactly, but they sit at different orbital indices and so map to different Pauli strings. The same reordering moves the top generator from D 4,5 $\rightarrow$ 12,13 to D 6,7 $\rightarrow$ 12,13. The two geometries are comparable as physics, their Pauli-level structures are not identical, and a difference of ten contexts carries no information.

9 The complete comparison

All four constructions are compared at one criterion: the winner must be separated from the rest of the pool at family-wise confidence $\delta = 0.05$. M1 and the baseline reach that with a single common radius gap/2 over the whole pool; M2 and M3 reach it by elimination, resolving each generator only to $\Delta_i/2$. No free schedule or precision parameter remains in any column. The cost scales as the inverse square of the relevant deficit, so the gap column is the main driver of difficulty.

M1 is given the exact gap, and this favours it. A non-adaptive method cannot know gap before measuring anything, yet the target radius gap/2 is computed from the exact gradients. M1 is therefore charged the loosest radius that happens to suffice, the most generous reading available. On H₄ side 1.0 HF the M1/M2 ratio moves from 0.66 to 168 as the target radius gap/ d varies over $d = 1, 2, 4, 8, 16$, M1 costing 561,216, 2,244,862, 8,979,447, 35,917,787 and 143,671,148 shots. The same oracle knowledge sets the good-enough tolerance of Sec. 12.5. Read every M1 column as *M1 at its best*.

The winners are D 2,3 $\rightarrow$ 10,11 (LiH), D 2,3 $\rightarrow$ 6,7 (both H₄ HF rows), D 0,1 $\rightarrow$ 6,7 (both H₄ CISD rows), D 4,5 $\rightarrow$ 12,13 (H₂O equilibrium) and D 6,7 $\rightarrow$ 12,13 (H₂O stretched), with $|g_{(1)}|$ of 0.243187, 0.295316, 0.142467, 0.315036, 0.098961, 0.304910, 0.006925, 0.388493 and 0.106873 down the rows. All four constructions return the true maximiser in all nine cases, which in oracle mode is a consistency check rather than a result: the rule reads exact gradients, so it cannot eliminate the maximiser. Whether it is still found once the gradients are estimated is the subject of Sec. 12.

Table 3: Oracle context-shot comparison at the winner-identification criterion, $\delta = 0.05$. $M1/M2 > 1$ means M2 is cheaper than M1; $M2/M3 > 1$ means M3 is cheaper than M2. Entries below unity are bold. Distances in Å; the baseline is the no-sharing diagnostic of Sec. 10. *Gap to four significant figures; full values follow the table.

Case	State	Gap*	Baseline	M1	M2	M3	M1/M2	M2/M3	M1/M3
LiH, $R=3.0$	HF	0.0964	6,819,980	3,563,904	1,140,583	638,423	3.12	1.79	5.58
H ₄ , side 1.0	HF	0.0215	4,848,846	2,244,862	854,629	566,951	2.63	1.51	3.96
H ₄ , side 1.0	CISD	0.1181	130,375	59,642	102,806	53,727	0.58	1.91	1.11
H ₄ , side 2.0	HF	0.0166	2,002,941	598,102	422,746	220,389	1.41	1.92	2.71
H ₄ , side 2.0	CISD	0.0519	180,604	58,068	86,857	33,432	0.67	2.60	1.74
H ₂ O eq., $R=0.9572$	HF	0.1011	204,022,319	172,147,028	25,096,606	22,130,852	6.86	1.13	7.78
H ₂ O eq., $R=0.9572$	CISD	0.00206	485,001,056,522	411,524,732,152	46,457,815,707	44,971,732,317	8.86	1.03	9.15
H ₂ O str., $R=1.75$	HF	0.0219	4,407,619,149	3,864,803,902	17,423,631	20,163,542	221.81	0.86	191.67
H ₂ O str., $R=1.75$	CISD	0.0170	6,317,143,181	5,715,348,735	168,305,383	165,843,674	33.96	1.01	34.46

M3 is cheaper than M1 in every row. M3 cheaper than M2 in eight rows of nine and M2 than M1 in seven, both subject to the routine-dependence of Sec. 14.

9.1 M3 is not guaranteed to beat M1

The intuition for $M_{M3} \leq M_{M1}$ is that M3 solves the same allocation over a shrinking subset of the constraints at radii never tighter than M1’s, so every breakpoint’s allocation costs less than M1’s in total. But the M3 cost is the sum over contexts of the per-context *maximum* across breakpoints, Eq. (14), and two breakpoints can load different contexts. **The ordering is not a theorem.** An adversarial search over small random problems — sparse fragment standard deviations and random gradients, hill-climbed on M_{M3}/M_{M1} — ends above unity from 19 of 60 unbiased starts and 24 of 60 starts biased towards a crowded top of the spectrum, reaching 1.09 in both. Every allocation in the worst instance agrees with a brute-force optimum to 10^{-4} , so the excess is real, not a solver artefact.

The mechanism is specific. In the worst instance, four arms over two contexts, the two runners-up are nearly tied, so the last two breakpoints differ in radius by 0.2%; but the arm that leaves between them was pinning the allocation to context A, and without it the leader’s optimum moves about seventy shots to context B. Taking the maximum per context pays for both loadings. The excess is therefore a property of the accounting — each breakpoint solved independently of the shots already committed — rather than of the method: topping up from the shots already taken instead gives 216.3 against M1’s 216.8 on the same instance. None of the nine cases comes close; the tightest is H₄ side 1.0 CISD at $M_{M1}/M_{M3} = 1.12$, whose top of the spectrum is also the most crowded in the set.

9.2 What sets the M1 cost

M1 must reach its common radius for every generator simultaneously, so its cost is set by the pool member with the largest reconstruction variance $\sum_{\alpha} \sigma_{i\alpha}^2$ — the *binding generator*, in the sense that its constraint in Eq. (21) is the one the final rescaling makes tight. Table 4 identifies it in every case.

Table 4: The generator whose constraint binds M1, against the winner. $\sum_{\alpha} \sigma^2$ is the reconstruction variance at one shot per parent context. The last column is the factor by which M1’s cost falls if it is handed the set of non-zero-gradient generators for free — information no non-adaptive method has.

Case	Binding generator	$ g $ of it	$\sum_{\alpha} \sigma^2$	same, winner	M1 all / M1 non-zero
LiH, $R=3.0$, HF	D 0,1 $\rightarrow$ 4,9	0	11.76	0.072	2.28
H ₄ , side 1.0, HF	D 0,1 $\rightarrow$ 4,7	0	2.713	0.447	1.71
H ₄ , side 1.0, CISD	D 0,1 $\rightarrow$ 6,7	1.4×10^{-1}	1.741	1.741	1.56
H ₄ , side 2.0, HF	D 0,1 $\rightarrow$ 4,7	0	0.218	0.067	1.83
H ₄ , side 2.0, CISD	S 1 $\rightarrow$ 7	1.9×10^{-31}	0.144	0.122	1.93
H ₂ O eq., HF	S 0 $\rightarrow$ 10	7.1×10^{-14}	537.4	1.317	2.20
H ₂ O eq., CISD	S 0 $\rightarrow$ 10	1.6×10^{-5}	530.8	1.284	2.01
H ₂ O str., HF	S 0 $\rightarrow$ 10	1.8×10^{-8}	531.6	0.592	2.35
H ₂ O str., CISD	S 0 $\rightarrow$ 10	2.0×10^{-4}	455.9	0.565	2.02

In six of the nine cases the binding generator has a gradient below the 10^{-6} threshold, and its reconstruction variance dwarfs the winner’s — by a factor of 164 on LiH and of 400 to 900 on H₂O. In the remaining three it carries a small but non-zero gradient, and on H₄ side 1.0 with the CISD state it is the winner itself.

This is not a defect of the comparison: M1 is non-adaptive, so it must bound every generator tightly enough to rule it out, and paying for high-variance operators that turn out not to matter is precisely the inefficiency elimination removes. Nor is it the whole story. Restricting M1 to the non-zero-gradient generators — the last column of Table 4 — cuts its cost by only 1.6 to 2.4, because the rest also carry substantial variance. The cost is spread over the pool rather than concentrated in one irrelevant operator. This is also a rough guide to how much Part III symmetry filtering could do for M1, although not a bound: filtering and tapering transform the operators

as well as dropping generators, so the two are not the same operation.

9.3 Where the cost actually sits: breadth and depth

The shot cost of a BAI method is not concentrated in the final elimination, as a naive reading of the $1/r^2$ scaling suggests. It splits into two terms, and which one dominates changes with the system.

One definition, applied the same way to both methods. Let $r_0 = \frac{1}{2} \max_i \Delta_i$ be the coarsest radius of the trajectory, the radius at which the first generator is eliminated and at which every generator is still active. *Breadth* is the cost of that first sweep,

$$M^{\text{breadth}} = \text{cost of resolving every generator to } r_0, \quad M^{\text{depth}} = M^{\text{total}} - M^{\text{breadth}}, \quad (23)$$

and the *breadth share* is $M^{\text{breadth}}/M^{\text{total}}$. For M2 the cost is separable, so $M^{\text{breadth}} = \sum_i (z_\delta S_i^{\text{own}}/r_0)^2$, writing $S_i^{\text{own}} = \sum_\alpha \sigma_{i\alpha}$ over generator i 's own contexts and S_i^{par} for the same sum over the parent contexts. For M3 it is the total of the allocation at the first breakpoint, before any elimination. Because shots are reusable, breadth is a cost neither method can avoid: it is the price of learning which generators are worth pursuing.

Table 5: Breadth share, Eq. (23): the fraction of each method's total that is already committed at the coarsest radius, when every generator is still active. The remainder is the depth term. Both columns use the same definition; M3's is the cumulative allocation at the first breakpoint of its elimination history.

Case	M2 breadth share	M3 breadth share	$\max_i \Delta_i/\text{gap}$
H ₄ , side 1.0, HF	3.0%	2.1%	13.7
H ₄ , side 1.0, CISD	87.1%	76.3%	1.2
H ₄ , side 2.0, HF	1.3%	0.8%	18.9
H ₄ , side 2.0, CISD	57.1%	47.7%	1.9
LiH, $R=3.0$, HF	93.9%	87.7%	2.5
H ₂ O eq., HF	89.4%	85.5%	3.0
H ₂ O eq., CISD	92.8%	81.3%	3.4
H ₂ O str., HF	80.1%	60.7%	17.8
H ₂ O str., CISD	94.5%	86.8%	6.3

The two HF H₄ rows are strongly depth-dominated: only 1 to 3% of the cost is committed before the first elimination, and essentially all of it goes into separating the leading pair. LiH, both H₂O geometries and the two H₄ CISD rows are breadth-dominated, with 57 to 94% of the cost already spent when every generator is still in play. What distinguishes them is $\max_i \Delta_i/\text{gap}$, the dynamic range the schedule has to cover: 13.7 and 18.9 on the two depth-dominated H₄ HF rows against 1.2 to 3.4 on H₄ CISD, LiH and H₂O equilibrium. Where the leading pair is nearly tied relative to the spread of the field, depth dominates; where the deficits are spread more evenly, breadth does. H₂O stretch, at 17.8 and 6.3, does not fit that reading, so the correlation is suggestive rather than established.

The two terms reward sharing very differently. In the breadth term a shared context serves the entire pool, 92 or 140 generators at once, and M3's advantage over M2 is correspondingly large. In the depth term it serves only the generators that survive to the final radius. M3's breadth share is below M2's in every one of the nine rows, which is what sharing is supposed to do: it makes the coarse sweep cheaper relative to the rest.

9.4 What decides the depth term

Two quantities decide it. Generator i 's own FC grouping is $\mathcal{M}_{i,1}, \dots, \mathcal{M}_{i,K_i}$; write $F_{i\alpha}$ for the part of C_i landing in parent context $\mathcal{M}_\alpha$. The *fragmentation penalty*

$$f_i = \left(\frac{\sum_\alpha \sigma_{i\alpha} \text{ over parent contexts}}{\sum_k \sigma_{ik} \text{ over its own } \mathcal{M}_{i,k}} \right)^2 \quad (24)$$

is the factor by which the shared grouping makes one gradient more expensive to resolve alone. It exceeds unity in every case here because Eq. (18) depends on $(\sum_\alpha \sigma_{i\alpha})^2$ rather than $\sum_\alpha \sigma_{i\alpha}^2$, so finer splitting raises the sum of standard deviations — but this is an *observation*, not a theorem: the parent grouping is not a refinement of the own grouping, since a parent context can merge Paulis the own grouping separates, so a larger context count does not by itself imply $f_i \geq 1$. The second quantity is $|\mathcal{A}| = |\{i : \Delta_i \leq \text{gap}\}|$, the number of generators still active at the final radius; by construction $|\mathcal{A}| \geq 2$.

In Table 6, K^{own} counts groups in the winner's own FC grouping and depends only on $\text{supp}(C_i)$, not the state; K^{parent} counts parent contexts in which the winner has a *non-zero* fragment standard deviation, which is state-dependent — hence 858 against 887, and 875 against 899, on the H₂O HF and CISD rows for the same winner. Contexts with $\sigma_{i\alpha} = 0$ contribute nothing and are not counted.

The scaling looks asymmetric. $|\mathcal{A}|$ is set by how many generators are nearly tied at the top, which has nothing to do with pool size and is two or three in every case here. The fragmentation penalty is larger on the larger

Table 6: Fragmentation of the winning gradient against the number of finalists. K^{own} is the number of groups in the winner’s own FC grouping; K^{parent} is the number of parent contexts in which it has a non-zero fragment standard deviation, which is why it depends on the state. Both are defined in the text above.

Case	State	K^{own}	K^{parent}	f	$ \mathcal{A} $	M2/M3
H ₄ , side 1.0	HF	10	22	1.13	2	1.51
H ₄ , side 1.0	CISD	7	28	1.07	3	1.91
H ₄ , side 2.0	HF	10	22	1.19	3	1.92
H ₄ , side 2.0	CISD	7	28	1.22	2	2.60
LiH, $R=3.0$	HF	58	520	3.49	3	1.79
H ₂ O eq., $R=0.9572$	HF	57	858	2.54	2	1.13
H ₂ O eq., $R=0.9572$	CISD	57	887	2.44	2	1.03
H ₂ O str., $R=1.75$	HF	50	875	3.62	2	0.86
H ₂ O str., $R=1.75$	CISD	50	899	2.94	2	1.01

systems, because a larger universal support spreads each individual gradient over more parent contexts: 22 on H₄, 520 on LiH, roughly 875 on H₂O. If that persisted, M3’s benefit in the depth term would have a ceiling while its cost in that term did not, and a crossover would follow.

This is a hypothesis, not a demonstration. f is not monotone in system size — LiH, the smaller system, gives 3.49 against 2.54 for H₂O at equilibrium — four systems cannot separate size from geometry and state, and f grows far more slowly than the raw context count (3.6 against 17.5 on H₂O stretch), because most parent contexts hold a negligible sliver of any one gradient. The mechanism is real but weak, and nothing here fixes where or whether a crossover occurs. It is a statement about the planning bound, and within it about the depth term only; Sec. 12.1 measures realised cost under a schedule, which the bound does not model, so the two do not compete.

9.5 The finalists do not share Pauli terms

The premise of universal grouping is that gradient operators share Pauli products. Across the pool they do: each Pauli product in $\mathcal{B}_0$ appears in roughly four different gradient commutators.

Table 7: Support sharing. Average reuse is $\sum_i |\text{supp}(C_i)|/|\mathcal{B}_0|$. Finalist overlap is the Jaccard overlap $|\text{supp}(C_{(1)}) \cap \text{supp}(C_{(2)})|/|\text{supp}(C_{(1)}) \cup \text{supp}(C_{(2)})|$ between the supports of the two leading generators. Supports do not depend on the state, but the identity of the finalists does, so the two H₄ rows differ. The remaining four cases have not been evaluated.

Case	$ \mathcal{B}_0 $	$\sum_i \text{supp}(C_i) $	Average reuse	Finalist overlap	M2/M3
H ₄ , side 1.0, HF	1,276	4,148	3.25	0.0%	1.51
H ₄ , side 1.0, CISD	1,276	4,148	3.25	27.8%	1.91
LiH, $R=3.0$, HF	30,196	120,384	3.99	35.1%	1.79
H ₂ O eq., $R=0.9572$, HF	81,566	314,720	3.86	0.0%	1.13
H ₂ O str., $R=1.75$, HF	81,566	314,720	3.86	0.0%	0.86

The two columns tell opposite stories. Pool-wide reuse of about four is why M3’s breadth term is efficient. But in three of the five cases the two finalists share *no* Pauli products at all, so at the depth stage there is nothing to share and M3 pays the fragmentation penalty for nothing.

The overlap column does not order the M2/M3 column across systems. The two cases with substantial finalist overlap, LiH and H₄ CISD, are cases where M3 keeps a clear advantage, and the H₂O rows with none are where it disappears — but H₄ HF has zero overlap and still gives 1.51, well above both H₂O rows, and its fragmentation penalty is six times smaller than LiH’s. Low finalist overlap is a plausible reason for M3 to lose its depth-stage advantage and is present where the advantage is smallest; five rows are not enough to make it *the* explanation.

9.6 Rebuilding the grouping once the field narrows

If the fixed parent contexts are the problem at the depth stage, the obvious remedy is to abandon them once the active set is small and build a fresh fully commuting grouping over the survivors’ union support. Phase one measures on the parent contexts while more than a chosen number of generators remain active; phase two regroups from scratch, using the same greedy first-fit routine and ordering.

The earlier shots are not lost when this happens. They are taken in different circuit settings, so no further shot can be added to the parent-context estimate of a survivor, but that estimate remains unbiased at reconstruction variance $V_i^{\text{par}} = \sum_{\alpha} \sigma_{i\alpha}^2/n_{\alpha}$, and the regrouped data give an independent one. The inverse-variance combination is unbiased at $V^{-1} = (V^{\text{par}})^{-1} + (V^{\text{new}})^{-1}$, so phase two need only supply the residual precision, arm by arm:

$$\frac{1}{\epsilon_i^2} = \frac{1}{\epsilon^2} - \frac{1}{V_i^{\text{par}}}, \quad (25)$$

imposed by scaling row i of the new fragment standard deviations by $1/\epsilon_i$ and solving Eq. (21) at unit epsilon. An arm whose parent data already meet the target drops out of phase two entirely. Switch points $|\mathcal{A}| \leq 2, 3, 4, 6, 10, 20$

and every breakpoint of the trajectory were tested on each case, both with this credit and with phase two charged the full cost of reaching the final radii.

Table 8: Staged regrouping against the better of M2 and M3, with the earlier shots credited by Eq. (25) and with them discarded. The switch point is the active-set size at which the parent contexts are abandoned.

Case	Better of M2, M3	Discard	Ratio	Reuse	Ratio
H ₄ , side 1.0, HF	566,951	502,858	0.89	494,591	0.87
H ₄ , side 1.0, CISD	53,402	53,402	1.00	50,029	0.94
H ₄ , side 2.0, HF	220,388	182,709	0.83	181,899	0.83
H ₄ , side 2.0, CISD	33,432	33,432	1.00	30,469	0.91
LiH, R=3.0, HF	639,465	639,464	1.00	597,169	0.93

With the credit, staged regrouping pays on all five cases, by 6 to 17%. Without it, three of the five show no gain at all and the procedure degenerates to M3. The saving is modest either way, and Table 5 says why: regrouping can only improve the depth term, and on the systems where M3 struggles the cost is dominated by breadth, which the remedy does not touch.

The credit buys robustness more than it buys shots. With the earlier shots discarded, the penalty for switching at the wrong point is severe — the total runs from 1.02 to 1.57 of the better method across switch points on LiH and on H₄ side 1.0 CISD, and the best switch point is early on H₄ and late on LiH. With the credit, the total is flat to within about 0.02 over a wide range and never exceeds 0.99. The choice of switch point, which otherwise has to be tuned per system with no rule that transfers, stops mattering.

Where the leverage is. It is at the first substantial elimination, not at the final pair. On H₄ side 1.0 HF the best switch is the first breakpoint, where sixteen of twenty-six generators leave at once and regrouping the ten survivors needs twelve contexts against fifty-three; the union support collapses disproportionately because those sixteen zero-gradient generators hold 708 of the 1,276 Pauli products in $\mathcal{B}_0$ (55.5%) and their supports are *disjoint* from the ten non-zero ones. The same pattern holds on LiH, where the best switch is $|\mathcal{A}| = 34$ out of 92.

This rules out an appealing alternative trigger. Since Sec. 9.5 shows the finalists often share no Pauli products at all, one might regroup at the moment the active supports decouple. That criterion fires on only one of the five cases, at $|\mathcal{A}| = 2$, and there it gives 0.95 against the 0.87 available from switching early; on the other four the active support never fully decouples and it never fires. Overlap remains high — 0.73 to 0.78 of the active support is still shared — exactly where the gain is largest, so high overlap and high leverage coincide rather than trade off. The switch should be triggered by the first large drop in $|\mathcal{A}|$, which is also where Part III symmetry filtering would act, without measuring anything first.

10 Attribution: the no-sharing baseline

The three methods are two binary choices, sharing and elimination, and they span only three of the four combinations. The fourth is a baseline that does neither: every generator is measured in its own fully commuting grouping, and every one of them is resolved to the same common radius used by M1. It is reported here as a diagnostic outside the method hierarchy, and it serves two purposes. It makes the two mechanisms separately attributable, and it is a sanity bound, since measuring every generator separately is always available as a fallback.

Table 9: Improvement each mechanism buys against the no-sharing baseline of Table 3. Sharing only is baseline/M1, elimination only baseline/M2, both baseline/M3. The product column is what “both” would be if the mechanisms acted independently; the interaction is the ratio of the two.

Case	State	Sharing only	Elimination only	Both	Product	Interaction
LiH, R=3.0	HF	1.91	5.98	10.68	11.44	0.93
H ₄ , side 1.0	HF	2.16	5.67	8.55	12.25	0.70
H ₄ , side 1.0	CISD	2.19	1.27	2.43	2.77	0.88
H ₄ , side 2.0	HF	3.35	4.74	9.09	15.87	0.57
H ₄ , side 2.0	CISD	3.11	2.08	5.40	6.47	0.84
H ₂ O eq., R=0.9572	HF	1.19	8.13	9.22	9.63	0.96
H ₂ O eq., R=0.9572	CISD	1.18	10.44	10.78	12.30	0.88
H ₂ O str., R=1.75	HF	1.14	252.97	218.59	288.50	0.76
H ₂ O str., R=1.75	CISD	1.11	37.53	38.09	41.49	0.92

Two things follow, and both bear on how the project should be described.

On the planning bounds the mechanisms are substitutes. The interaction is below unity in every one of the nine cases, between 0.57 and 0.96: applying both buys less than the product of applying each alone, because elimination removes precisely the situation that sharing exploits, many generators active at once. **That reading holds only for the bound.** The bound resolves every M3 arm to the common radius, and Sec. 11 shows that on LiH and H₂O the shared allocation resolves the leader far more tightly than that; with M3 eliminating on its

actual precision the interaction exceeds unity on all five of those rows. There the mechanisms compound. On H_4 they remain substitutes.

Sharing contributes less than elimination, and less on the larger systems. On its own, universal grouping is worth a factor of 2.2 to 3.4 on H_4 , 1.9 on LiH, and only 1.1 to 1.2 on H_2O . Elimination on its own is worth between 1.27 and 252.97: the low end is the H_4 side 1.0 CISD row, where the gap is wide and there is little for elimination to save, and the high end is the H_2O stretch HF row. The two ranges overlap, so “sharing contributes little” should be read as *relative to elimination* and not as an absolute statement — a factor of 3.4 is a real saving on its own terms. What the table does establish is the ordering of the two mechanisms and its direction: the sharing factor is smallest on the largest system in this set, on all four H_2O rows. Whether that is a size effect or a property of these particular molecules cannot be settled from four systems.

11 The M3 bound undercounts shared precision

Every M3 number so far resolves each active arm to the common radius r and removes arm i once $2r < \Delta_i$. That is not what the allocation delivers. A shared allocation is sized by its binding arm, so every other active arm is resolved more tightly than required, and the parent contexts keep every shot already taken, so the shots spent separating arms that have since left keep sharpening the survivors. A run that eliminates on each arm’s actual reconstruction variance $V_i = \sum_{\alpha} \sigma_{i\alpha}^2 / n_{\alpha}$ — as any real run computing its own variances would — drops arm i as soon as some j satisfies

$$|g_j| - z_{\delta} \sqrt{V_j} > |g_i| + z_{\delta} \sqrt{V_i}. \quad (26)$$

With $n_{\alpha}(r) = \max(n_{\alpha}^{\text{so far}}, n_{\alpha}^{(A)}(r))$ every V_i is non-increasing as r falls, so the dominance margin is monotone in r and each elimination radius is found by bisection. The trajectory is followed event by event with no schedule, and cost is charged per context at its largest demand, as in Eq. (14). It is the $\gamma \rightarrow 1$ limit of the noiseless trial of Sec. 12, and agrees with it to within one schedule step wherever that trial resolves (a test pins the agreement). M2 needs no such column: each M2 arm is resolved to exactly the common radius, so its bound already uses its actual precision. M1 and the baseline eliminate nothing.

Table 10: M3 on the common-radius planning bound and eliminating on actual precision, Eq. (26). *Breadth* is M3’s breadth share from Table 5. The interaction is that of Table 9, recomputed with the actual-precision M3; values above unity, where the two mechanisms compound rather than substitute, are set in bold.

Case	State	Breadth	M3			M2/M3		Interaction	
			Bound	Actual	Ratio	Bound	Actual	Bound	Actual
LiH, $R=3.0$	HF	87.6%	639,465	333,272	0.52	1.78	3.42	0.93	1.79
H_4 , side 1.0	HF	2.1%	566,951	569,020	1.00	1.51	1.50	0.70	0.70
H_4 , side 1.0	CISD	76.7%	53,403	49,729	0.93	1.93	2.07	0.88	0.95
H_4 , side 2.0	HF	0.8%	220,389	228,733	1.04	1.92	1.85	0.57	0.55
H_4 , side 2.0	CISD	47.7%	33,432	31,553	0.94	2.60	2.75	0.84	0.89
H_2O eq., $R=0.9572$	HF	85.5%	22,133,048	13,246,592	0.60	1.13	1.89	0.96	1.60
H_2O eq., $R=0.9572$	CISD	81.2%	45,017,730,601	24,416,305,313	0.54	1.03	1.90	0.88	1.61
H_2O str., $R=1.75$	HF	60.7%	20,163,542	14,581,169	0.72	0.86	1.19	0.76	1.05
H_2O str., $R=1.75$	CISD	86.9%	165,495,967	122,188,158	0.74	1.02	1.38	0.92	1.25

The bound is conservative on LiH and H_2O and accurate on H_4 . Eliminating on actual precision costs M3 0.52 to 0.74 of its planning bound on all five LiH and H_2O rows, and 0.93 to 1.04 on all four H_4 rows. Where it comes out slightly above the bound, the event-driven trajectory separates eliminations that the bound lumps into one breakpoint, and the per-context maxima of Eq. (14) then pay a little for the extra allocations, as in Sec. 9.1.

The saving comes from the leader. A losing arm is removed once the leader leads it by the sum of their two actual radii, where the bound charges both at r . At the breadth breakpoint, where most of M3’s cost on these systems sits, the allocation is sized by the losers that are expensive to resolve; if the leader’s own fragments are cheap by comparison it is resolved far below the target. Its standard error is then a fraction κ of the target: 0.48 on LiH and 0.56 on H_2O at equilibrium, against 0.94 to 1.00 on every H_4 row, where the leader is itself binding. A binding loser is then released at $2/(1 + \kappa)$ times the bound’s radius — 1.35 on LiH, about 1.8 times fewer shots for the stage that carries 88% of the cost. On H_2O the losers are over-resolved as well (median κ of 0.70 to 0.78 against 1.00 on LiH), which widens the release radius further. This is why breadth share alone does not predict the saving: H_4 side 1.0 CISD is 77% breadth but its leader binds, and it gains 7%.

Three conclusions change. First, *the M2-versus-M3 comparison widens on every LiH and H_2O row*: to 3.42 on LiH and 1.19 to 1.90 on H_2O , including stretched H_2O HF, where the bound’s 0.86 becomes 1.19 and stays above unity under every grouping routine (Sec. 14). Second, *the LiH finite-shot differential of Sec. 12.1 is accounted for*: the realised ratio of 3.04 sits between the bound’s 1.78 and the actual-precision 3.42, noise costing M3 slightly more than it costs M2. Third, *the two mechanisms compound on LiH and H_2O* . The interaction against the no-sharing

baseline, below unity on every row at bound level, becomes 1.05 to 1.79 on all five of those rows and stays at 0.55 to 0.95 on H_4 . Sharing spills precision onto arms the allocation was not sized for, and that precision is exactly what lets elimination fire early. The substitution reading of Sec. 10 is a property of the common-radius bound, and holds for M3 as it would actually run only where the leader binds.

12 Finite-shot behaviour

Every number above is an oracle planning bound. The elimination rule reads exact gradients, so it cannot misfire, and the family-wise confidence is asserted rather than measured. This section replaces the exact gradients with estimates built from a finite number of shots and runs the same elimination rule on those.

Protocol. A real run does not know the deficits in advance and cannot jump to the exact breakpoints, so each trial follows a geometric radius schedule: the common radius starts at $\max_i |g_i|$, wide enough that no arm is eliminated for free, and is multiplied by $\gamma = 0.5$ each round. At each round the allocation of Eq. (21) is solved for the current active set and radius, and each context’s shot count is raised to that value if it is higher, never lowered. Each context’s fluctuation is carried as a running sum, each round adding the draw for its new shots only, so $\hat{g}_i$ is distributed as $\mathcal{N}(g_i, \sum_{\alpha} \sigma_{i\alpha}^2/n_{\alpha})$ at the accumulated n_{α} and successive rounds are correlated as real accumulated data would be. The elimination rule of Eq. (9) is then applied. A trial ends when one arm remains or the radius falls below $10^{-3} \max_i |g_i|$. Every case uses 200 trials at seed 0; the trial count is set so that the standard error on a mean cost is a few per cent, the resolution the comparisons need. The sweeps below use 250, 300 and 500 trials: those tables report *rates* rather than costs, and the counts were raised to tighten the Wilson intervals, which at 300 and 500 trials reach $\pm 1.3\%$ and $\pm 0.8\%$ at zero observed errors. The starting radius and the floor are both taken from the exact $\max_i |g_i|$ and are therefore oracle quantities, though mild ones: a real run would take the starting radius from a cheap pilot estimate of the largest gradient, and the floor only prevents a non-terminating run. This is a weaker use of oracle knowledge than M1’s exact gap (Sec. 9), but it is not none: the best-arm methods discover their *stopping* point from the data, not their starting point.

The inflation is mostly the protocol, not noise. Running the identical trial loop with the noise draws set to zero isolates what the schedule and the stopping rule cost on their own, at matched stopping rules, and the remainder is what noise adds (Table 15). Schedule-only inflation is 1.74 to 3.48 for M2 and 0.88 to 2.85 for M3, while noise multiplies it by only 0.84 to 1.40 and 0.87 to 2.67. Several noise factors fall below unity, a lucky fluctuation ending a run a round early. M3’s noiseless trial cost falls *below* its planning bound on three rows, as low as 0.88, because stopping when one arm remains is a weaker criterion than resolving the leader to gap/2. The “inflation” column therefore mixes a weaker stopping criterion, which lowers cost, with discretisation and noise, which raise it.

Both confounds are removed in Sec. 12.1, by sweeping the shrink factor towards the exact limit and by matching the stopping rule to the planning criterion. The result is that the inflation is overwhelmingly discretisation: at $\gamma = 0.95$ it falls to 1.03–1.35 for M2 and 0.73–1.29 for M3. The M2/M3 differential visible in the table below survives on LiH and both H_2O CISD rows, and Sec. 11 shows it belongs to the bound.

The two deliberate approximations, the normal limit and independence between generators, are listed in Sec. 7; Sec. 15 removes the second.

Columns. *Bound* is the oracle planning cost of Table 3; *Infl.* is the mean total shots over the 200 trials divided by it; *Correct* is the fraction of trials returning the true maximum-gradient generator. M2/M3 columns are ratios of the corresponding entries, standard errors combined in quadrature.

Table 11: Realised cost against the planning bound, 200 trials per row at seed 0, all uncertainties one standard error of the mean. Every row selected the true maximiser in all 200 trials, a Wilson 95% interval of [0.981, 1]; the *Correct* column is therefore constant and omitted. The non-adaptive baseline and M1 realise their bound exactly, having no schedule to overshoot, and are omitted.

Case	State	M2: BAI-FC-IG		M3: BAI-FC-UG		M2/M3	
		Bound	Infl.	Bound	Infl.	Bound	Realised
LiH, $R=3.0$	HF	1,140,583	3.16 ± 0.06	638,423	1.40 ± 0.03	1.79	4.03 ± 0.12
H_4 , side 1.0	HF	854,629	2.21 ± 0.08	566,951	2.33 ± 0.10	1.51	1.43 ± 0.08
H_4 , side 1.0	CISD	102,806	2.99 ± 0.05	53,727	2.74 ± 0.04	1.91	2.09 ± 0.04
H_4 , side 2.0	HF	422,746	2.33 ± 0.05	220,389	2.48 ± 0.05	1.92	1.81 ± 0.05
H_4 , side 2.0	CISD	86,857	2.77 ± 0.06	33,432	2.95 ± 0.09	2.60	2.43 ± 0.09
H_2O eq., $R=0.9572$	HF	25,096,606	3.02 ± 0.06	22,130,852	1.53 ± 0.04	1.13	2.24 ± 0.07
H_2O eq., $R=0.9572$	CISD	46,457,815,707	3.02 ± 0.07	45,017,730,601	1.36 ± 0.03	1.03	2.30 ± 0.07
H_2O str., $R=1.75$	HF	17,423,631	2.75 ± 0.06	20,163,542	1.66 ± 0.04	0.86	1.43 ± 0.04
H_2O str., $R=1.75$	CISD	168,305,383	2.95 ± 0.07	165,495,967	2.07 ± 0.05	1.02	1.45 ± 0.05

Realised means are Bound $\times$ Infl.; the largest is 1.4×10^{11} shots for M2 on H_2O CISD at equilibrium and the smallest 9.9×10^4 for M3 on H_4 side 2.0 CISD.

The two H₂O CISD rows carry the largest bounds in the set, up to 4.6×10^{10} shots for M2 at equilibrium; the surrogate samples estimators rather than outcomes, so shot totals of that size cost nothing to simulate.

The scatter is not negligible and is the reason the errors are quoted. Repeated runs of the same configuration at different seeds move a realised total by 2 to 4%, which is comparable to some of the differences between adjacent rows and larger than the difference between the bound and realised M2/M3 ratios on the H₄ HF rows.

12.1 The inflation is the radius schedule

The trials above use a geometric schedule of shrink factor $\gamma = 0.5$. A real run does not know the deficits Δ_i in advance, so it discovers them by shrinking a common radius, and a coarse schedule overshoots every elimination point. Sweeping γ separates that overshoot from noise: as $\gamma \rightarrow 1$ the schedule approaches the exact-limit trajectory the planning bound assumes, so whatever inflation survives is not discretisation. A second control replaces the trial’s stopping rule — halt when one arm remains — with the criterion the bound assumes, halt at radius gap/2; the default rule is the weaker of the two and can stop before the bound’s terminal radius.

Table 12: Realised cost divided by planning bound, at the coarse schedule and near the exact limit, with the realised M2/M3 ratio against its bound-level value. 200 trials per entry (100 on LiH) at seed 0, one-arm stopping rule; the matched rule changes no entry by more than 0.1 except where noted in the text.

Case	M2 inflation		M3 inflation		realised M2/M3		bound M2/M3
	$\gamma=0.5$	0.95	$\gamma=0.5$	0.95	$\gamma=0.5$	0.95	
H ₄ , side 1.0, HF	2.21	1.03	2.33	1.12	1.43	1.38	1.51
H ₄ , side 1.0, CISD	2.99	1.35	2.76	1.29	2.09	2.02	1.93
H ₄ , side 2.0, HF	2.33	1.10	2.48	1.17	1.81	1.80	1.92
H ₄ , side 2.0, CISD	2.77	1.21	2.95	1.19	2.43	2.64	2.60
LiH, $R=3.0$, HF	3.01	1.27	1.40	0.74	3.84	3.04	1.78
H ₂ O eq., CISD	3.02	1.29	1.36	0.73	2.30	1.81	1.03
H ₂ O str., CISD	2.95	1.23	2.07	0.91	1.45	1.38	1.02

Almost all of the inflation is the schedule. At $\gamma = 0.95$ M2 costs 1.03 to 1.35 of its bound and M3 0.73 to 1.29, against 2.2 to 3.0 at $\gamma = 0.5$. The coarse schedule was measuring its own overshoot. A deployment would tune γ , and the cost of not tuning it is a factor of two to three.

On H₄ the realised comparison reproduces the bound. The realised M2/M3 ratio at $\gamma = 0.95$ is 1.38, 2.02, 1.80 and 2.64 against bound-level 1.51, 1.93, 1.92 and 2.60, and the matched stopping rule moves these to 1.43, 2.03, 1.87 and 2.54. The bound-level comparison of Sec. 9 therefore carries over to finite shots on all four rows.

LiH does not, and the reason is the bound, not the noise. Its realised ratio is 3.04 under the one-arm rule and 3.12 under the matched rule, against a bound of 1.78, and neither moves with γ : M2 inflates to 1.27 while M3 falls to 0.74 of its own bound. The same trial run with every noise draw set to zero already costs M3 only 0.55 of its bound at $\gamma = 0.95$, with every arm leaving at a radius above its exact $\Delta_i/2$; noise then raises M2 and M3 by similar factors, 1.26 and 1.38. So the trial is not beating the bound through lucky draws. It eliminates on each arm’s actual precision, which on a shared allocation is far better than the common radius the bound assumes. Sec. 11 quantifies this for every case; on LiH it accounts for the whole differential.

12.2 The comparison with M1 under finite shots

M1 and the no-sharing baseline are non-adaptive: they solve one allocation at one radius and realise their planning bound exactly, with no schedule to overshoot. The best-arm methods do not, so comparing realised cost against M1’s bound is sensitive to γ in a way Table 3 is not.

Table 13: M1 against the *realised* cost of the best-arm methods at $\gamma = 0.5$. M1’s bound is its realised cost. Entries below unity, where the best-arm method is more expensive than static all-gradient estimation, are set in bold.

Case	State	M1	M1 / M2 realised	M1 / M3 realised
LiH, $R=3.0$	HF	3,563,904	0.99	3.99
H ₄ , side 1.0	HF	2,244,862	1.19	1.70
H ₄ , side 1.0	CISD	59,642	0.19	0.40
H ₄ , side 2.0	HF	598,102	0.61	1.09
H ₄ , side 2.0	CISD	58,068	0.24	0.59
H ₂ O eq., $R=0.9572$	HF	172,147,028	2.27	5.07
H ₂ O str., $R=1.75$	HF	3,864,803,902	80.7	115.3
H ₂ O eq., $R=0.9572$	CISD	411,524,732,155	2.93	6.74
H ₂ O str., $R=1.75$	CISD	5,715,348,740	11.5	16.7

At this schedule M3 loses to M1 on both H₄ CISD rows, which it won on the bounds. **That reordering does not survive a finer schedule.** At $\gamma = 0.95$ M3 realises 39,840 on H₄ side 2.0 CISD against M1’s 58,068, and with the matched stopping rule it realises 54,872 on H₄ side 1.0 CISD against M1’s 59,642: M3 beats M1 on all nine

rows. The inversions in the table are the $\gamma = 0.5$ overshoot, not a property of finite sampling, and the bound-level claim that M3 is cheaper than M1 stands under finite shots as well.

What the table does show is where the margin is thin. Elimination pays where M1 is expensive — H_2O , where the factor stays between 2 and 115 — and barely pays where M1 is already cheap, which is the regime conclusion 4 identifies. M2, which had already lost both CISD rows at bound level, also loses H_4 side 2.0 HF and draws level on LiH.

12.3 The one inversion does not survive

On H_2O stretch with the Hartree–Fock reference the planning bounds give $\text{M2}/\text{M3} = 0.86$, the only row in which M3 is more expensive than M2. Under finite shots the same row gives 1.43 ± 0.04 , and M3 wins. The inversion belongs to the bound: the trial eliminates on each arm’s actual precision, the bound does not, and with M3 eliminating on actual precision the ratio is 1.08 to 1.52 across all five grouping routines (Sec. 14), where the bound itself moves over 0.86 to 1.05. On actual precision M3 is the cheaper method on this row under every routine tested.

12.4 The confidence convention is far stricter than δ , by an amount the data can only bound

No trial in the main finite-shot study of Table 11 selected the wrong generator, and sweeping the family-wise failure probability over nearly its whole range barely changes that. Errors do appear elsewhere — at $\delta = 0.90$ and 0.99 below, and at small z_δ in the lower block of Table 14 — so the statement is about this sweep, not about every run in the report.

How much stricter cannot be measured here. With 300 trials and zero errors a Wilson 95% interval puts the rate anywhere between 0 and 1.3%. The 1.2×10^{-5} quoted below is the analytic prediction of Eq. (27), not a measurement; resolving a rate that small needs of order 10^6 trials. What is established is that the realised rate is at most 1.3% where 0.05 was asked for, about four times smaller, and that it barely moves as δ is swept — enough for the practical conclusion that δ is not the knob, not enough to say by how much the convention over-delivers.

The same holds on two further cases. Repeating the sweep over $\delta \in \{0.05, 0.5, 0.9, 0.99\}$ with 300 trials per entry, LiH returns no wrong generator for either method at any δ , and H_4 side 2.0 HF none at $\delta = 0.05$ and at most 1.7% [0.7, 3.8] at $\delta = 0.9$ — a nominal family-wise failure probability of ninety per cent delivering under two per cent. The rate is set by the stopping radius and the gap, as below, and the confidence level barely enters.

The reason is structural. Whether a run halts on one surviving arm or on the radius, the leader can only be separated from the runner-up once the common radius has come down to about $\text{gap}/2$, at which point each estimate has standard error $\text{gap}/(2z_\delta)$. The leader’s margin is one gap, so in units of the standard deviation of the difference it is $z_\delta\sqrt{2}$, and

$$P(\text{wrong selection}) \approx 2\bar{\Phi}(z_\delta\sqrt{2}), \quad (27)$$

where $\bar{\Phi}$ is the upper-tail standard normal function. The single-competitor probability is $\bar{\Phi}(z_\delta\sqrt{2})$; the factor of two is a conservative allowance, not a derivation, covering the comparison being between absolute values and more than one arm sitting near the top. Dropping it halves every predicted rate and changes no conclusion. The Bonferroni term enters only through z_δ , and the stopping radius supplies most of the protection.

Table 14: Confidence sweeps on H_4 side 1.0 HF. Upper block: the stated δ against the realised selection-error rate, 300 trials per entry. Lower block: the confidence factor set directly, 500 trials per entry. Brackets are Wilson 95% intervals. At these trial counts none of the upper rows distinguishes a rate of 10^{-5} from one of 10^{-2} .

δ	z_δ	M2 error [95% CI]	M3 error [95% CI]	M2 shots	M3 shots
0.05	3.10	0.0% [0, 1.3]	0.0% [0, 1.3]	1,891,179	1,324,843
0.40	2.42	0.0% [0, 1.3]	0.0% [0, 1.3]	1,232,695	849,844
0.90	2.11	0.7% [0.2, 2.4]	0.3% [0.1, 1.9]	1,128,410	691,975
0.99	2.07	0.0% [0, 1.3]	0.3% [0.1, 1.9]	969,099	741,804
—	1.39	12.2% [9.6, 15.4]	10.6% [8.2, 13.6]	1,586,040	989,854
—	1.80	1.8% [1.0, 3.4]	2.0% [1.1, 3.6]	1,113,341	819,684
—	2.10	0.4% [0.1, 1.5]	0.0% [0, 0.8]	1,211,688	685,567
—	3.10	0.0% [0, 0.8]	0.0% [0, 0.8]	2,004,737	1,335,779

Eq. (27) predicts 0.049, 0.011, 0.003 and 1.2×10^{-5} at $z_\delta = 1.39, 1.80, 2.10$ and 3.10. It accounts only for the final comparison and ignores generators wrongly eliminated earlier in the run, so it underestimates the error at small z_δ : it predicts 5% at $z_\delta = 1.39$ where the simulation gives ten to twelve. The cost is also not monotone. Below $z_\delta \approx 1.8$ it rises again, because bad eliminations prolong the run. There is an optimum near $z_\delta \approx 1.8$ to 2.1, worth about 1.7 in realised cost at an error rate whose 95% interval reaches 3.6% at worst — a real saving, but not the factor of five the bound suggested. Both conventions are selectable in the implementation; the tables in this report retain the Bonferroni one. The sweep was run on one case, H_4 side 1.0 with the Hartree–Fock reference, and

the optimum has not been checked on any other, so $z_\delta \approx 1.8$ to 2.1 should be treated as a single-system observation rather than a recommended default.

12.5 Good-enough selection

A run does not have to identify the maximiser; it is often enough to return a generator that is not much worse. Fix a tolerance $\tau > 0$ — written τ to keep it distinct from the deficits Δ_i — and stop at the first round t at which every surviving candidate is certified within τ of the best,

$$\max_{j \in \mathcal{A}_t} (|\hat{g}_j| - r_j) \geq \max_{i \in \mathcal{A}_t} (|\hat{g}_i| + r_i) - \tau, \quad (28)$$

returning $\arg \max_{i \in \mathcal{A}_t} |\hat{g}_i|$. The run is counted a success if the returned generator’s true absolute gradient is within τ of the true maximum.

On H_4 side 1.0 HF at $\tau = \text{gap}/4 = 5.38 \times 10^{-3}$, over the same 200 trials at seed 0 as Table 11, M2 falls from $1,889,539 \pm 64,528$ to $1,398,814 \pm 32,966$ shots and M3 from $1,320,830 \pm 57,944$ to $904,382 \pm 20,201$, savings of 26% and 31%. In all 200 trials the generator returned was still the exact maximiser. Over 300 trials each, the same tolerance saves 18% (M2) and 24% (M3) on H_4 side 2.0 HF but only 2 to 3% on LiH, in every trial returning a generator within τ of the best. The saving tracks how crowded the top of the spectrum is: $\tau = \text{gap}/4$ spares the final tightening, which is most of the cost where the leading gradients are close together and almost none of it where the gap is wide. Note that $\tau = \text{gap}/4$ uses the exact gap, so this inherits the oracle-knowledge caveat of Sec. 9; a deployment would set τ from a chemical accuracy target instead.

13 Parameter sensitivity

13.1 Removing the two free parameters

Two arbitrary constants have been removed. M1 is evaluated at a target radius of $\text{gap}/2$, the loosest common radius at which a non-adaptive estimate separates the winner from the field, so that all three methods answer the same question. The BAI methods are evaluated in the schedule-free limit of Sec. 5 rather than on a geometric schedule, because a coarse schedule overshoots each elimination point and overpays.

Neither change is cosmetic. Under $\gamma = 0.5$, M3 costs more than M1 on three H_4 rows. In the exact limit it does not, on any row (Sec. 9), so that ordering is a property of the limit rather than of the method; under a coarse schedule it can fail, and Sec. 12.2 shows the same failure appearing and then disappearing under finite shots as γ is refined.

Table 15: Decomposition of the finite-shot cost at matched stopping rules. *Schedule* is the identical trial loop with the noise draws set to zero, divided by the planning bound; *noise* is the realised mean divided by that; their product is the realised inflation of Table 11. The last three columns track the M2/M3 ratio through the same three stages.

Case	State	M2		M3		M2/M3		
		Schedule	Noise	Schedule	Noise	Bound	Schedule	Realised
LiH, $R=3.0$	HF	2.25	1.40	0.88	1.59	1.79	4.56	4.03
H_4 , side 1.0	HF	1.74	1.27	1.62	1.44	1.51	1.62	1.43
H_4 , side 1.0	CISD	3.48	0.86	2.16	1.27	1.91	3.09	2.09
H_4 , side 2.0	HF	2.75	0.85	2.85	0.87	1.92	1.85	1.81
H_4 , side 2.0	CISD	2.28	1.21	1.11	2.67	2.60	5.36	2.43
H_2O eq., $R=0.9572$	HF	3.37	0.90	0.88	1.74	1.13	4.34	2.24
H_2O str., $R=1.75$	HF	3.26	0.84	1.91	0.87	0.86	1.48	1.43

Bold marks rows where the noiseless trial costs *less* than the planning bound. Under the planning stopping rule instead, the same schedule overpays by 1.64 to 3.80 and moves the M2/M3 ratio by only -4% to $+20\%$; the contrast with the Schedule column measures how much the stopping rule, rather than the shrink factor, is doing. Refining the schedule converges onto the exact limit as it must: Refining the schedule converges onto the exact limit as it must: on H_4 side 1.0 HF, M2 falls $1,494,274 \rightarrow 1,267,755 \rightarrow 1,080,113 \rightarrow 885,115$ and M3 $928,366 \rightarrow 872,427 \rightarrow 712,219 \rightarrow 584,142$ at $\gamma = 0.5, 0.7, 0.85, 0.97$, against exact limits of 854,629 and 566,951. `verify_part1.py` asserts this convergence.

For M2 the limit is the closed form

$$M_{\text{M2}} = \sum_i \left(\frac{z_\delta \sum_\alpha \sigma_{i\alpha}}{\Delta_i/2} \right)^2, \quad (29)$$

with Δ_i replaced by the gap for the leader. For M3 the trajectory has one breakpoint per distinct Δ_i and the allocation of Eq. (21) is evaluated at each.

13.2 The confidence level does not change the comparisons

Every planning bound scales with z_δ^2 , so every ratio in Table 3 is invariant under the family-wise failure probability. Verified for $\delta \in \{0.2, 0.05, 0.01, 0.001\}$: the H_4 side 1.0, HF ratios stay at 2.63 and 1.51 to better than one part in 10^3 , while the absolute costs move by about a factor of two.

This invariance is a property of the bounds and does not extend to realised cost. Once a schedule and finite shots are introduced, z_δ changes where each elimination fires and therefore how much each method overshoots, and the two methods are not affected equally: Table 14 gives realised M2/M3 ratios of 1.60, 1.36, 1.77 and 1.50 at $z_\delta = 1.39, 1.80, 2.10$ and 3.10 on one H_4 row — a spread of 1.3, with no monotone pattern. What δ does to the realised error rate is treated in Sec. 12.4.

14 Routine independence

Every number above comes from one first-fit grouping with one insertion order. First-fit is deterministic given an order, but the order is a free convention, so the conclusions have to be tested against it. The insertion order is a parameter: three deterministic conventions — by decreasing weight (**weight**, the default used everywhere else), by increasing weight (**reverse-weight**) and lexicographic — together with a **random** family under seeds 0, 1, 2, 3. The H_4 rows use all seven, LiH the three deterministic orderings and **random** seed 0, the H_2O rows the three deterministic orderings and **random** seeds 0, 1. The ordering is applied to every grouping the methods use — the parent contexts of M1 and M3 and the individual groupings of M2 and the baseline — so each row recomputes all four constructions from scratch. Every combination is recorded per case in ***_grouping_routines.csv**.

Table 16: Spread of the method ratios across groupings of the same problem, each recomputing the baseline and all three methods. The last column records whether the ordering of the methods is the same under every grouping tested.

Case	Contexts	M1/M2	M2/M3	M1/M3	Ordering fixed
H_4 , side 1.0, HF	49–61	1.91–2.63	1.51–1.97	3.35–4.55	yes
H_4 , side 1.0, CISD	49–61	0.40–0.58	1.93–2.78	1.10–1.15	yes
H_4 , side 2.0, HF	49–61	1.21–1.54	1.92–2.56	2.61–3.33	yes
H_4 , side 2.0, CISD	49–61	0.44–0.72	2.56–3.92	1.71–1.91	yes
LiH, $R=3.0$, HF	980–1,127	2.58–3.12	1.79–2.11	5.22–5.58	yes
H_2O eq., HF	2,366–2,849	5.63–6.86	1.13–1.36	7.65–7.78	yes
H_2O eq., CISD	2,366–2,849	7.20–8.86	1.03–1.34	9.14–9.63	yes
H_2O str., HF	2,356–2,870	181–250	0.86–1.05	172–262	no
H_2O str., CISD	2,356–2,870	27.4–38.0	0.94–1.24	33.9–35.7	no

On seven of the nine rows the ratios move by at most a factor of 1.64 and the ordering of the methods is identical under every grouping tested. Those rows are routine-independent for the quantities in the table.

The two stretched H_2O rows are the exceptions, and the only ones. Their M2/M3 ratios run over 0.86 to 1.05 (HF) and 0.94 to 1.24 (CISD), spanning unity: which of the two best-arm methods is cheaper on the planning bound is itself a function of an arbitrary convention. At equilibrium the same five orderings give 1.13 to 1.36 and 1.03 to 1.34, entirely above unity, so both equilibrium rows *are* resolved in M3’s favour; on each, the value quoted in Table 3 is the least favourable of the five. The two geometries differ here for the same reason their Pauli supports differ (Sec. 8). The M1 versus M3 comparison is never in doubt on any H_2O row, at 7.7 to 262 across the sweep. The stretched CISD row’s canonical-routine ratio reads 1.02 in the sweep against 1.01 in Table 3 for the reason given in Sec. 5.

On actual precision both stretched rows are resolved. Repeating the sweep with M3 eliminating on each arm’s actual precision (Sec. 11) gives M2/M3 of 1.08 to 1.52 on stretched H_2O HF and 1.35 to 2.36 on stretched H_2O CISD, across the same five orderings: M3 is the cheaper method under every one. The irresolution above is a property of the common-radius bound, which charges M3 for precision its allocation delivers anyway; it is not a property of the method.

15 Within-context correlation

Assumption (iii) of Sec. 7 is removed here: two generators read off the same shots in a shared context, so their estimates are correlated, and modelling that correlation makes the simulation exact for M1 and M3 as well.

The context covariance. For each context α , let $c_{i\alpha} = (F_{i\alpha} - \langle F_{i\alpha} \rangle) |\psi\rangle$ be the centred fragment vector. The covariance is the Gram matrix

$$\text{Cov}(F_{i\alpha}, F_{j\alpha}) = \Re \langle c_{i\alpha} | c_{j\alpha} \rangle, \quad (30)$$

which is positive semidefinite by construction and whose diagonal is exactly $\sigma_{i\alpha}^2$. Both properties are load-bearing and neither is automatic. The polarisation identity $\text{Cov}(A, B) = \frac{1}{2}[\text{Var}(A+B) - \text{Var}(A) - \text{Var}(B)]$ is equivalent in exact arithmetic, but a parent context is dominated by symmetry-forbidden fragments, for which $|\psi\rangle$ is an eigenstate of $F_{i\alpha}$ and the variance is zero; there the identity subtracts three quantities that are individually at the

floating-point noise floor. Computing each variance as $\langle F^2 \rangle - \langle F \rangle^2$ compounds the problem, since that form also cancels catastrophically for a near-eigenstate fragment. Together the two produced correlations as large as 136 and context matrices with relative eigenvalues as low as -6×10^{264} . Eq. (30) with $\text{Var}(F) = \|c\|^2$ removes both cancellations: measured over all 53 parent contexts of the H_4 rows the worst relative eigenvalue is -7×10^{-16} , no pair violates Cauchy–Schwarz, and every diagonal matches $\sigma_{i\alpha}^2$ to 10^{-15} . The test suite pins all three properties (Sec. 19); the consequence of losing them is recorded at the end of this section.

15.1 The covariance-aware elimination rule

Where the *marginal* rule of Eq. (9) compares $|\hat{g}_i|$ and $|\hat{g}_j|$ at their individual radii, the *pairwise* rule compares each pair at the radius of the combination the comparison actually depends on. Because the comparison is between absolute values, that combination is $\hat{g}_i - s_{ij}\hat{g}_j$ with $s_{ij} = \text{sign}(\hat{g}_i\hat{g}_j)$, so i is eliminated when some active j satisfies

$$|\hat{g}_j| - |\hat{g}_i| > z_\delta \sqrt{V_{ij}}, \quad V_{ij} = \sum_{\alpha} \frac{1}{n_{\alpha}} \text{Var}(F_{i\alpha} - s_{ij}F_{j\alpha}), \quad (31)$$

with the same z_δ of Eq. (20). When the two fragments are correlated in the direction that helps, V_{ij} is below $\text{Var}(\hat{g}_i) + \text{Var}(\hat{g}_j)$ and the rule fires earlier.

The sign factor is required. Using $\text{Var}(\hat{g}_i - \hat{g}_j)$ unconditionally is not only the wrong combination for an opposite-sign pair; it is not well defined, because the sign of a gradient depends on the arbitrary phase of the Hartree–Fock orbital it excites from. Only $s_{ij} \text{Cov}(\hat{g}_i, \hat{g}_j)$ is phase-invariant, and Eq. (31) uses exactly that.

Table 17: M3 under correct joint noise, 250 trials per row at seed 0. Uncertainties are one standard error; brackets are Wilson 95% intervals on the correct-selection rate.

Case	Noise model, rule	Shots	Correct [95% CI]	vs indep.
H_4 , side 1.0, HF	independent, marginal	$1,334,743 \pm 51,258$	100.0% [98.5, 100]	—
	correlated, marginal	$1,323,818 \pm 51,343$	100.0% [98.5, 100]	0.99
	correlated, pairwise	$685,298 \pm 23,477$	100.0% [98.5, 100]	0.51
H_4 , side 1.0, CISD	independent, marginal	$148,219 \pm 1,989$	100.0% [98.5, 100]	—
	correlated, marginal	$151,312 \pm 1,604$	100.0% [98.5, 100]	1.02
	correlated, pairwise	$93,530 \pm 2,619$	100.0% [98.5, 100]	0.63
H_4 , side 2.0, HF	independent, marginal	$544,228 \pm 10,397$	100.0% [98.5, 100]	—
	correlated, marginal	$522,022 \pm 11,231$	100.0% [98.5, 100]	0.96
	correlated, pairwise	$366,027 \pm 14,242$	100.0% [98.5, 100]	0.67

Modelling the correlation changes the cost by 1 to 4% on all three rows. The covariance-aware rule then cuts M3’s realised cost by 33 to 49% at 100.0% [98.5, 100] correct selection throughout. This is the mechanism Part II [6] proposes for the decisive stage, appearing as a direct reduction in shots rather than as the outcome of an optimisation over coefficient splitting.

Why the construction of Eq. (30) matters. With the covariance built from the polarisation identity instead, H_4 side 2.0 HF returned 75.6% [69.9, 80.5] correct under the marginal rule and 51.2% [45.0, 57.3] under the pairwise rule, at eleven to seventeen times the cost. The mechanism is specific: an indefinite context matrix makes the accumulated V_{ij} of Eq. (31) negative, the implementation clamps it at zero before taking the square root, and the arm is then eliminated on any lead whatever. On that case every trial hit a negative V_{ij} ; in a representative one the true leader was removed at round three against a lead of 1.8×10^{-2} at radius zero, from $\text{Var}_i = 6.4 \times 10^{-4}$, $\text{Var}_j = 4.7 \times 10^{-4}$ and a reported covariance of 2.1×10^{-3} , an implied correlation of 3.8. The failure is loudest where the runner-up is exactly degenerate — D 0,3 $\rightarrow$ 4,7 and D 1,2 $\rightarrow$ 5,6 both at $|g| = 0.298387$, so $|A| = 3$ — and the gap is only 5% of $|g_{(1)}|$. It is recorded here because a negative variance silently clamped to zero is not a visible failure, and because the two properties that exclude it, positive semidefiniteness and an exact diagonal, are cheap to test and were not being tested.

A diagnostic, defined once. For an active pair write

$$R_{ij} = \frac{\text{Var}(\hat{g}_i - s_{ij}\hat{g}_j)}{\text{Var}(\hat{g}_i) + \text{Var}(\hat{g}_j)} = 1 - \frac{2 s_{ij} \text{Cov}(\hat{g}_i, \hat{g}_j)}{\text{Var}(\hat{g}_i) + \text{Var}(\hat{g}_j)}, \quad (32)$$

the factor by which the decisive comparison is harder ($R_{ij} > 1$) or easier ($R_{ij} < 1$) than the marginal radii assume. Because s_{ij} and Cov change sign together under an orbital phase flip, R_{ij} is phase-invariant. Over the three cases it is 0.929, 1.084 and 1.168.

16 Conclusions from the current data

Table 18 gives the evidence level; items resting on one system or on bounds alone should not be generalised without the work in Sec. 18.

1. The four constructions are implemented as separate modules saving their FC groups, fragment variances and elimination histories, evaluated at one criterion with no free schedule or precision parameter. The validation gate passes on all nine cases to 4×10^{-15} Ha.
2. **M3 is cheaper than M1 in every case**, by 1.1 to 192 on the planning bounds and on realised cost in all nine simulated rows. **This is not a theorem**: under the sum-of-maxima accounting of Eq. (14) small problems with a nearly tied runner-up give M_{M3}/M_{M1} up to 1.09 (Sec. 9.1). Every M1 column assumes M1 is handed the exact gap.
3. **M3 is cheaper than M2 on every row once it eliminates on actual precision**. On the planning bound M3 wins eight rows of nine, and both equilibrium H_2O rows are robust to the grouping routine (1.13–1.36 and 1.03–1.34); both stretched rows are not, their bound-level ratios spanning unity (0.86–1.05 and 0.94–1.24). With M3 eliminating on actual precision those two become 1.08–1.52 and 1.35–2.36 across the same routines, and the finite-shot trial agrees (1.43 and 1.45 at $\gamma = 0.5$). The bound-level irresolution is the bound’s (Secs. 14 and 11).
4. **M2 is not always cheaper than M1**: seven bound-level rows of nine, five of nine on realised cost at $\gamma = 0.5$. The counterexamples are wide-gap rows where static estimation is already cheap, but LiH has a comparably wide gap and does not invert, so the gap alone is not the explanation.
5. **On the planning bounds the two mechanisms are substitutes; as M3 would actually run, they compound on LiH and H_2O** . At bound level the interaction against the no-sharing baseline is 0.57 to 0.96 in all nine cases. With M3 eliminating on its actual precision it is 1.05 to 1.79 on all five LiH and H_2O rows and 0.55 to 0.95 on H_4 (Sec. 11).
6. **Elimination dominates in seven rows of nine**. Elimination alone is worth 1.27 to 252.97, sharing alone 1.11 to 3.35; sharing wins on both H_4 CISD rows. The sharing factor is smallest on all four H_2O rows, which with four systems is a correlation with system, not a trend with size, and refers to bounds only.
7. **The finite-shot inflation is the radius schedule, not noise**. At the coarse schedule $\gamma = 0.5$ both methods cost 2.2 to 3.0 times their bounds. Taking γ to 0.95 collapses that to 1.03–1.35 for M2 and 0.73–1.29 for M3, so what the coarse schedule measures is its own overshoot of each elimination point. The realised M2/M3 ratio then tracks the bound-level ratio on all four H_4 rows under both stopping rules. On LiH and both H_2O CISD rows it does not: 3.04, 1.81 and 1.38 against bounds of 1.78, 1.03 and 1.02. The trial eliminates on actual precision and the bound does not, and the actual-precision M3 of Sec. 11 puts those ratios at 3.42, 1.90 and 1.38, so the differential is the bound’s, not the noise’s.
8. **The confidence convention does not control the selection-error rate**. Sweeping δ from 0.05 to 0.99 moves the realised rate only within 0–1.7% on three cases, and at $\delta = 0.05$ 300 trials bound it below 1.3% on each. Inverting Eq. (27) for 5% gives 10 to 12% in simulation, and an optimum sits near $z_\delta \approx 1.8$ –2.1, both on H_4 side 1.0 HF. Good-enough stopping at $\tau = \text{gap}/4$ saves 18 to 31% where the top of the spectrum is crowded and 2 to 3% on LiH, always returning a generator within τ .
9. **Conclusions 2 to 6 are routine-independent at bound level on seven of the nine rows**: all four H_4 rows, both H_2O equilibrium rows and LiH, each recomputed under five to seven groupings of every kind the methods use. Ratios move by at most 1.64 and no ordering changes. The two stretched H_2O rows are the exceptions (conclusion 3). The finite-shot and correlated studies are unswept.
10. **Within-context correlation is a saving on all three cases tested**. Modelling it changes cost by 1 to 4%, and the covariance-aware pairwise rule then cuts M3’s realised cost by 33 to 49% at 100.0% [98.5, 100] correct selection throughout. The saving requires a context covariance that is positive semidefinite with an exact diagonal (Sec. 15).
11. **Regrouping once the field narrows is worth 6 to 17%**, on all five cases tested, provided the shots already taken are credited by inverse-variance combination rather than discarded. Discarding them leaves three of the five cases with no gain at all. The credit matters less for the size of the saving than for its robustness: with it the total is flat to within 0.02 across a wide range of switch points, without it the penalty for switching late reaches 1.57 (Sec. 9.6).
12. **The M3 planning bound is conservative on the larger systems**. Eliminating on actual precision, M3 costs 0.52 to 0.74 of its bound on all five LiH and H_2O rows and 0.93 to 1.04 on H_4 . The saving comes from the leader: where its fragments are cheap relative to the losers that size the breadth allocation, it is resolved to 0.48–0.79 of the target, and losers leave at up to 1.35 times the bound’s radius (Sec. 11). The finite-shot trial eliminates the same way, which is why M3 realises below its bound on LiH and H_2O .
13. All numbers are single-step, single-state diagnostics counting shots only — not a full ADAPT trajectory, and not circuit-level cost, where M2 and the baseline need 3.5 to 6.1 times as many distinct settings (Table 1).

Table 18: Evidence behind each conclusion. “Bound” is the oracle planning model, “finite-shot” the surrogate of Sec. 12, “swept” the number of rows on which the claim survives every grouping of Sec. 14.

#	Claim	Bound	Finite-shot	Swept
2	M3 < M1	9 of 9	9 of 9 at $\gamma = 0.95$	9 of 9
3	M3 < M2	8 of 9; 9 of 9 actual	9 of 9	7 of 9; 9 of 9 actual
4	M2 < M1	7 of 9	5 of 9 at $\gamma = 0.5$	9 of 9
5	Substitutes	9 of 9; 4 of 9 actual	—	no
6	Elimination dominates	7 of 9	—	9 of 9
7	Inflation is the schedule	—	7 cases, 2–5 values of γ	no
8	δ does not control error	—	3 cases, 300–500 trials	no
10	Covariance-aware rule	—	3 cases, 250 trials	no
11	Regrouping with reuse	5 cases	—	no
12	M3 bound conservative	9 cases	2 cases, noiseless trial	no
13	Distinct-setting counts	4 systems	—	no

17 Interface with Parts II and III

Part II improves M3 by using measured variances and covariances to optimise overlapping universal operators and coefficient splitting. Sec. 9.3 gives that work a sharper target, but Sec. 9.5 also raises a caution that should be settled before it is built.

The mechanism Part II [6] proposes for the decisive stage is a pairwise-variance identity: best-arm identification only requires the pairwise differences to be resolved, and, with the sign convention of Eq. (31),

$$\text{Var}(\hat{g}_i - s_{ij}\hat{g}_j) = \sum_{\alpha} \frac{1}{n_{\alpha}} \text{Var}(F_{i\alpha} - s_{ij}F_{j\alpha}), \quad (33)$$

so the covariance that helps enters through co-measurement inside a shared context, not through the two generators sharing Pauli products. That premise was tested on the finalist pair of each system, inside the parent contexts, by comparing the cost of resolving the difference alone against the cost of resolving both generators to the same radius.

Table 19: Separating the two finalists inside the shared parent contexts: resolving only their difference against resolving each of them, both to radius gap/2. All sums are over parent contexts, so these are the S^{par} of Sec. 9.3. The two finalist gradients have the *same* sign in all four cases ($s_{ij} = +1$), so the relevant observable is $C_i - C_j$ and no sign correction applies; the one pair that is opposite-sign under the orbital phases used here, H₄ side 2.0 HF, is not among them.

Case	S_i^{par}	S_j^{par}	S_{i-j}^{par}	Resolve both	Resolve difference
H ₄ , side 1.0, HF	1.921	1.731	2.455	554,857	499,910
LiH, $R=3.0$, HF	2.170	2.180	3.623	48,712	67,577
H ₂ O eq., $R=0.9572$, HF	7.263	12.173	17.803	1,002,360	1,581,034
H ₂ O str., $R=1.75$, HF	6.239	7.794	13.005	10,636,302	18,049,318

Resolving the difference is cheaper only on H₄, by 1.11; on LiH and both H₂O rows it is more expensive, by up to 1.7. Writing S for S^{par} : resolving two generators separately costs $S_i^2 + S_j^2$ in units of ϵ^{-2} against S_{i-j}^2 for the difference, so the difference wins only when $S_{i-j} < \sqrt{S_i^2 + S_j^2}$, which for comparable arms needs roughly 30% cancellation. The measured cancellation $1 - S_{i-j}/(S_i + S_j)$ is 33% on H₄, 17% on LiH and 7% on H₂O stretch. It does not track the finalist overlaps of Table 7 — H₄ has zero overlap and the largest cancellation, LiH 35% overlap and half the cancellation — consistent with the premise that the useful covariance comes from co-measurement rather than shared support, but leaving the size of the cancellation unexplained.

This does not refute Part II, whose other mechanisms — coefficient splitting over an overlapping cover, empirically estimated covariances, adaptive choice of the next context — are untouched. It does mean the pairwise-difference mechanism should not be assumed to pay, and that cancellation must be measured on the intended benchmark first. A second caution: that mechanism acts on the depth term, only 12% of M3’s cost on LiH and 39% on H₂O stretch, so its effect is bounded even where it pays.

Part III [7] changes the ADAPT trajectory so that the commutators first become symmetry-filtered and tapered. When Part III is active, the Part I object C_i is replaced by

$$C_{i,\text{tap}}^{\parallel,k} = \text{Taper}_{\mathcal{R}_k}(\Pi_{\mathcal{R}_k}([H, G_i])), \quad (34)$$

and then M1, M2, or M3 is applied to these reduced operators. Part I tables do not include this reduction.

18 Remaining work in Part I

1. **Adopt actual precision in the M3 accounting**, or report both columns. The planning bound is conservative by up to a factor of two on the larger systems (Sec. 11), and the attribution, the M2/M3 comparison and the staged regrouping of Sec. 9.6 were all computed on it; the last has not been repeated on actual precision.

2. **Allocate incrementally.** Solving each breakpoint independently of the shots already committed is what lets M3 exceed M1 (Sec. 9.1). Topping up from committed shots repairs the counterexample found; whether it guarantees $M_{M3} \leq M_{M1}$ in general is open.
3. **Extend the routine sweep** to the finite-shot and correlated studies, which are still run under one grouping routine.
4. **Price the excluded cost components.** Distinct settings differ by 3.5 to 6.1 times between the method families (Table 1), and no circuit depth is counted, so the M2-versus-M3 comparison is incomplete even where the shot counts are resolved.
5. **Shot-level sampling.** The surrogate samples estimators; neither its Gaussian marginals nor its joint draw has been compared with multinomial sampling of measurement outcomes.
6. **Establish whether the CISD state is a fair proxy** for a mid-trajectory ADAPT state, which needs trajectory data.

19 What is tested, and what is not

Covered. 41 unit tests. Against dense linear algebra, independently of the Pauli-string evaluator: every commutator expansion against $HG_i - G_iH$ as a matrix, every gradient against a central finite difference of the energy, and every fragment variance and context covariance against dense fragment matrices, all on H_4 side 1.0 HF. Conventions and closed forms: basis ordering and the big-endian convention; the qubit Hamiltonian against the PySCF RHF and FCI energies on every case, as a hard gate; the allocation against Eq. (18) for a single gradient; the fragment decomposition summing back to the gradient operator. Groupings: every parent context and every individual grouping is a partition of its support into fully commuting sets. Invariances and pins: method totals and pairwise variances unchanged under negating any subset of generators; both square- H_4 RHF energies; the context covariances positive semidefinite, within Cauchy–Schwarz, with exact diagonals and non-negative pairwise variances under either sign convention (Sec. 15); the counterexample to $M_{M3} \leq M_{M1}$ of Sec. 9.1; and the actual-precision M3 of Sec. 11 as the limit of the noiseless trial. `verify_part1.py` adds 68 checks on the published results: each reported total is the sum of the file that explains it, the gate passed, the allocation meets its confidence target, $M_{M3} \leq M_{M1}$ on every case, all methods agree on the selected generator, repeated runs agree exactly, the ratios are invariant to δ , and a geometric schedule converges onto the exact limit. All pass.

Not covered, in rough order of value: shot-level sampling of measurement outcomes, against which neither the Gaussian marginals (assumption (i) of Sec. 7) nor the joint draw has been compared; the dense checks on any system beyond H_4 ; KKT residuals for the allocation on systematic multi-constraint instances; and regression pinning of every table entry against a commit hash.

Square H_4 has more than one RHF solution, and the build pins one. From eight perturbed starting guesses at side 1.0 Å, seven converge to -1.761075054 Ha and one to -1.694889592 Ha; at side 2.0 Å all eight agree on the energy to 4×10^{-13} Ha but their densities differ by up to 1.0 in max-norm — degenerate symmetry partners, which relabel which generator carries which gradient. PySCF’s default guess is not a well-defined target here: with the same PySCF version, one numerical environment converges at side 1.0 to the lower solution and another to the higher, which is internally unstable and whose gradients differ by up to 0.25. The build therefore follows any internal instability until the stability analysis passes. This reaches -1.761075054 Ha in every environment tried, reproduces the reported gradients to 10^{-15} and the method totals exactly, and changes nothing on the other seven geometries, all of which are stable at the default guess. A test pins both H_4 energies.

Within one solution the statistics are phase-invariant: a sign flip changes g_i and the matching row and column of the covariance together, leaving $|g_i|$, every fragment variance and $s_{ij} \text{Cov}(\hat{g}_i, \hat{g}_j)$ — hence R_{ij} of Eq. (32) — unchanged, and a test confirms that negating an arbitrary subset of generators leaves every method total and every pairwise variance of Eq. (31) unchanged. *Across* solutions nothing is guaranteed, not even generator labelling, which is why signed gradients and raw covariances from different runs must not be compared.

The implementation should be regarded as validated at the level these tests reach and no further.

20 Reproduction

```
python -m pip install -r requirements.txt
python scripts/run_part1_methods.py --all --out runs
python scripts/run_finite_shot.py --case H4_square_eq_side1p0_HF \
    --trials 200 --out runs
python scripts/experiment_q7_decomposition.py --case LiH_R3p0_HF
python scripts/experiment_spillover.py --out runs
python scripts/experiment_grouping_routines.py --case H20_stretch_HF \
```

```

--out runs --resume
python scripts/experiment_m3ext.py --case H4_square_eq_side1p0_HF
python scripts/experiment_calibration.py --case LiH_R3p0_HF
python scripts/experiment_m3_vs_m1_adversarial.py --compressed
python -m pytest tests -q && python scripts/verify_part1.py --runs runs

```

Case identifiers accepted by `--case` are `LiH_R3p0_HF`, `H4_square_eq_side1p0_{HF,CISD}`, `H4_square_stretch_side2p0_{HF,CISD}`, `H2O_{eq,stretch}_{HF,CISD}`.

Each case writes a per-method summary together with the FC groups, fragment standard deviations and BAI histories behind it, and records the interpreter and package versions with every run. `verify_part1.py` checks the published results rather than the implementation, as listed in Sec. 19; all 68 checks pass alongside 41 unit tests.

Determinism and seeds. The planning bounds contain no randomness and re-run exactly; `verify_part1.py` asserts this. The H_2O sweep takes several minutes per grouping, and `--resume` saves each grouping as it completes, so an interrupted sweep loses at most the one in progress. The finite-shot tables use `numpy` default-RNG streams at seed 0, except Table 14, whose per-entry seeds are recorded in the run summaries. The routine sweep uses `random` orderings at seeds 0 to 3.

Numerical settings. PySCF defaults are used for the SCF convergence threshold and initial guess, except that square H_4 needs the phase convention noted in Sec. 19 to make covariances reproducible. The CISD state is the lowest eigenpair of the Hamiltonian restricted to the ≤ 2 -excitation space, taken with a dense symmetric eigensolver below 2^{12} basis states and with `scipy.sparse.linalg.eigsh` at tolerance 10^{-12} above it. The convex-solver cross-check of Sec. 5 used SLSQP on randomly generated instances with up to eight arms and twenty contexts, agreeing to better than one part in 10^6 .

Environment. Code and run outputs live at https://github.com/Zephrou5747/ADAPT_Sampling. Pin Python 3.10.12, PySCF 2.14.0, OpenFermion 1.8.1, NumPy 2.2.6 and SciPy 1.15.3; `requirements.txt` states minimum versions only and does not by itself recreate that environment. On one desktop core the wall clock is dominated by building the gradient problem: about 16 s for H_4 , 37 s for LiH, two to three minutes for H_2O ; each 200-trial run then takes 0.1 to 40 s per method.

References

- [1] H. R. Grimsley, S. E. Economou, E. Barnes, and N. J. Mayhall, “An adaptive variational algorithm for exact molecular simulations on a quantum computer,” *Nature Communications* **10**, 3007 (2019). <https://doi.org/10.1038/s41467-019-10988-2>
- [2] V. O. Shkolnikov, N. J. Mayhall, S. E. Economou, and E. Barnes, “Avoiding symmetry roadblocks and minimizing the measurement overhead of adaptive variational quantum eigensolvers,” *Quantum* **7**, 1040 (2023). <https://doi.org/10.22331/q-2023-06-12-1040>
- [3] P. G. Anastasiou, N. J. Mayhall, E. Barnes, and S. E. Economou, “How to really measure operator gradients in ADAPT-VQE,” *arXiv:2306.03227* (2023). <https://arxiv.org/abs/2306.03227>
- [4] R. Huang and A. F. Izmaylov, “Quantum Gambling: Best-Arm Strategies for Generator Selection in Adaptive Variational Algorithms,” *arXiv:2509.14917* (2025); *J. Chem. Theory Comput.* (2026). <https://arxiv.org/abs/2509.14917>
- [5] E. Rossi, E. R. Kjellgren, A. F. Izmaylov, S. P. A. Sauer, K. M. Ziem, and S. Coriani, “Resource-efficient energy-based operator selection in fermionic ADAPT-VQE via exact Hamiltonian transformation,” *arXiv:2606.04786* (2026). <https://arxiv.org/abs/2606.04786>
- [6] L. Wang, “Part II: Variance-Aware Universal Operators for Grouped Best-Arm Identification in ADAPT-VQE Gradient Measurements,” internal project note (2026); source in `reports/` of the repository cited in Sec. 20.
- [7] L. Wang, “Part III: Symmetry-Staged ADAPT-VQE with Residual Stabilizer Tapering,” internal project note (2026); source in `reports/` of the repository cited in Sec. 20.